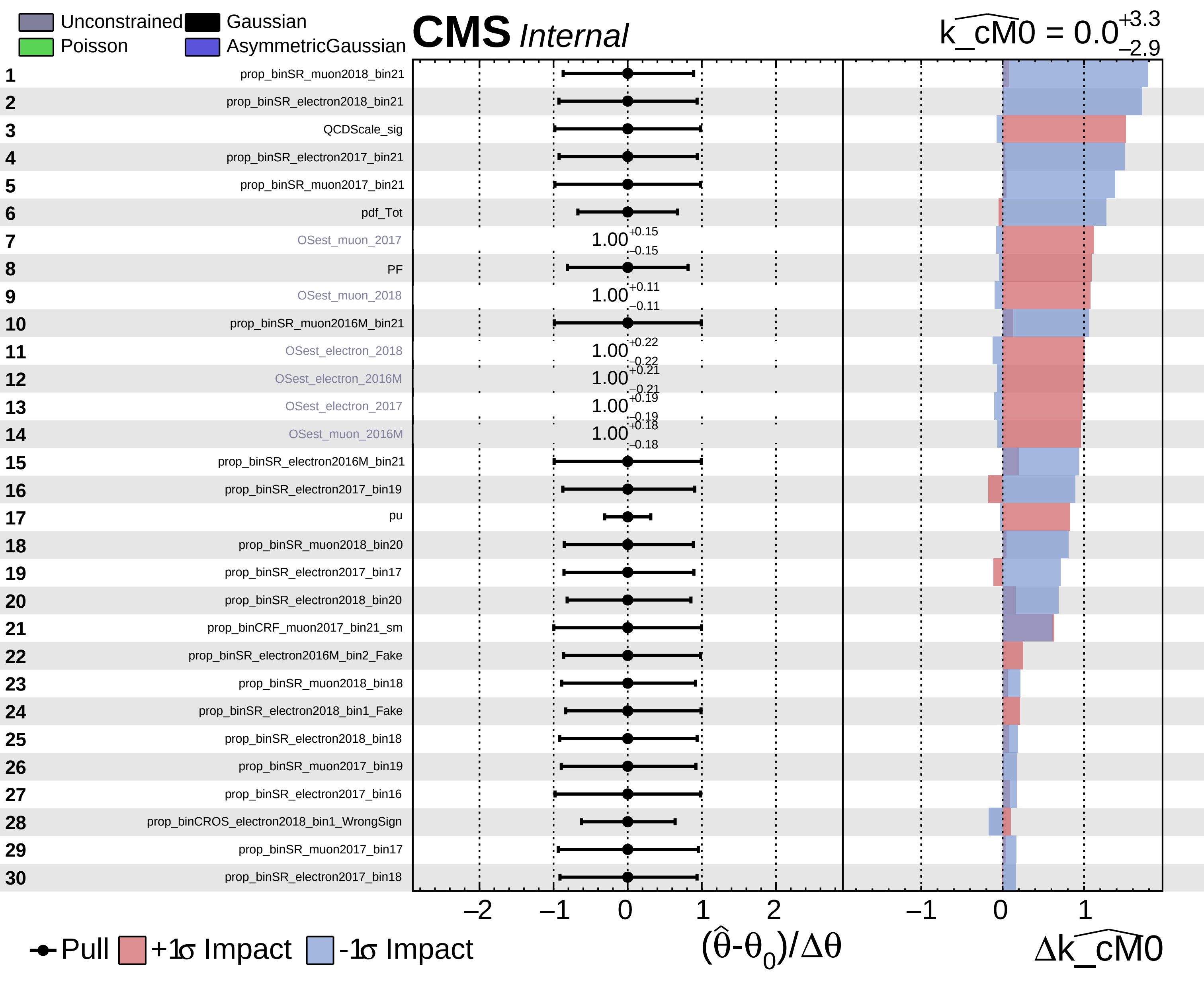
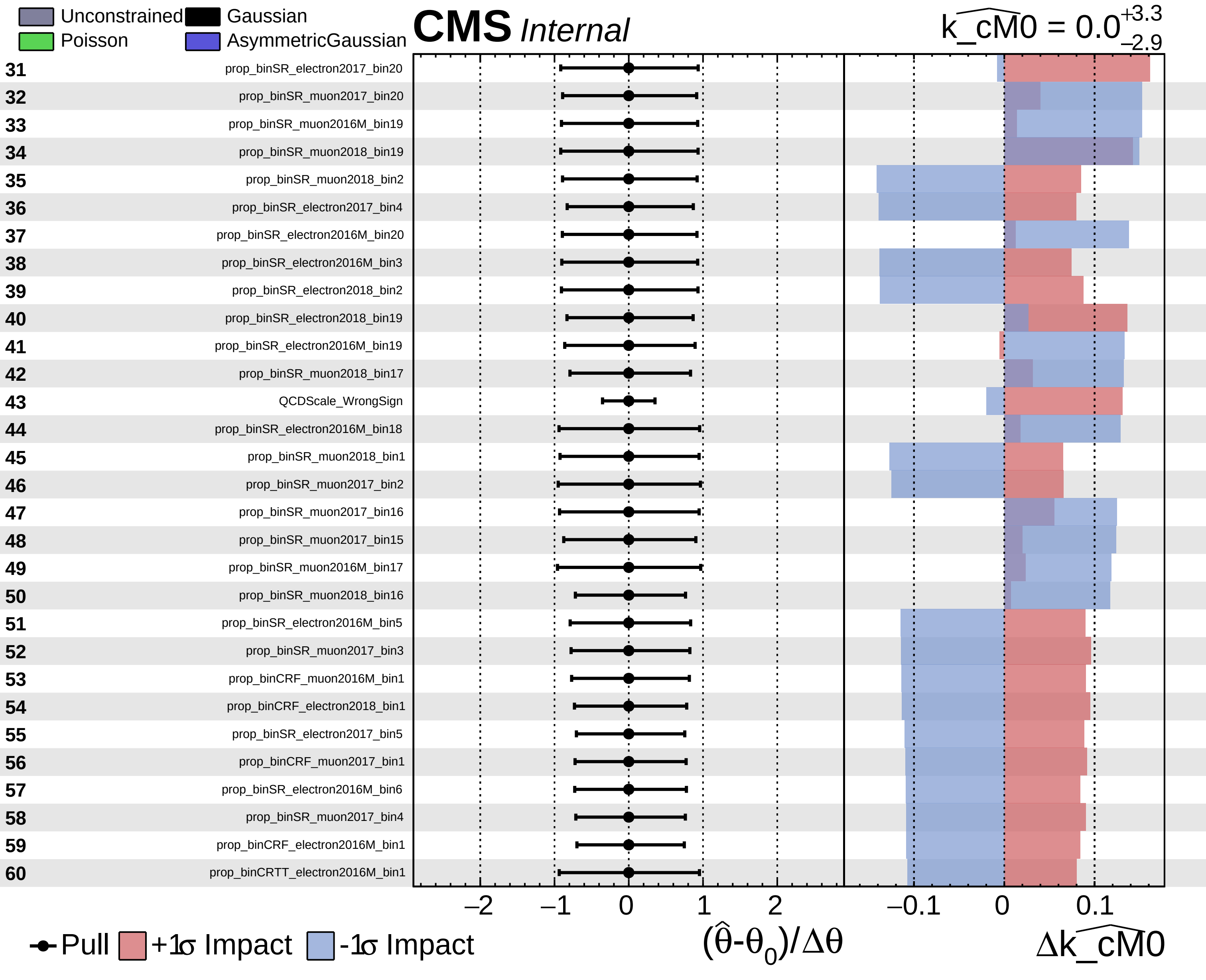
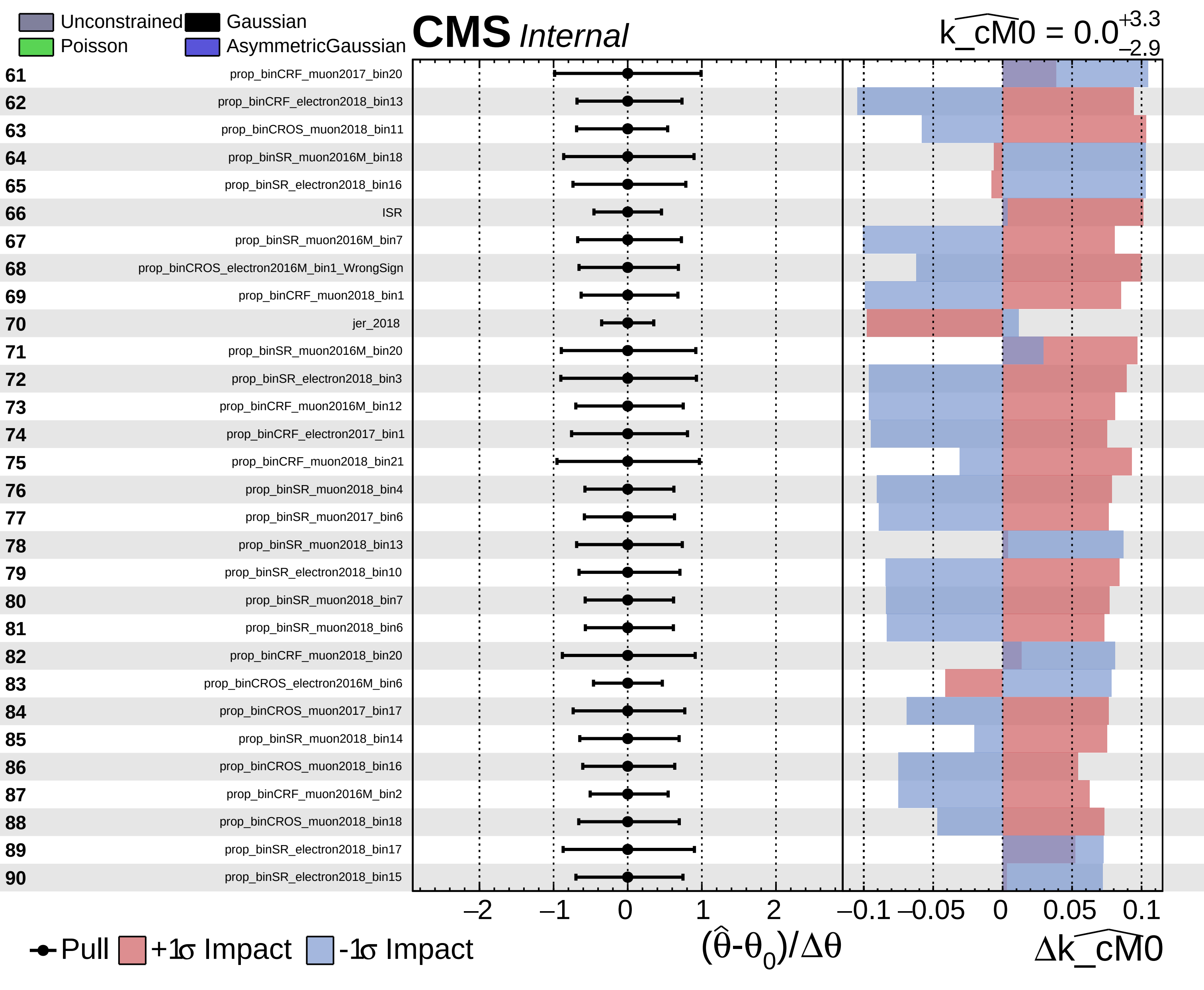
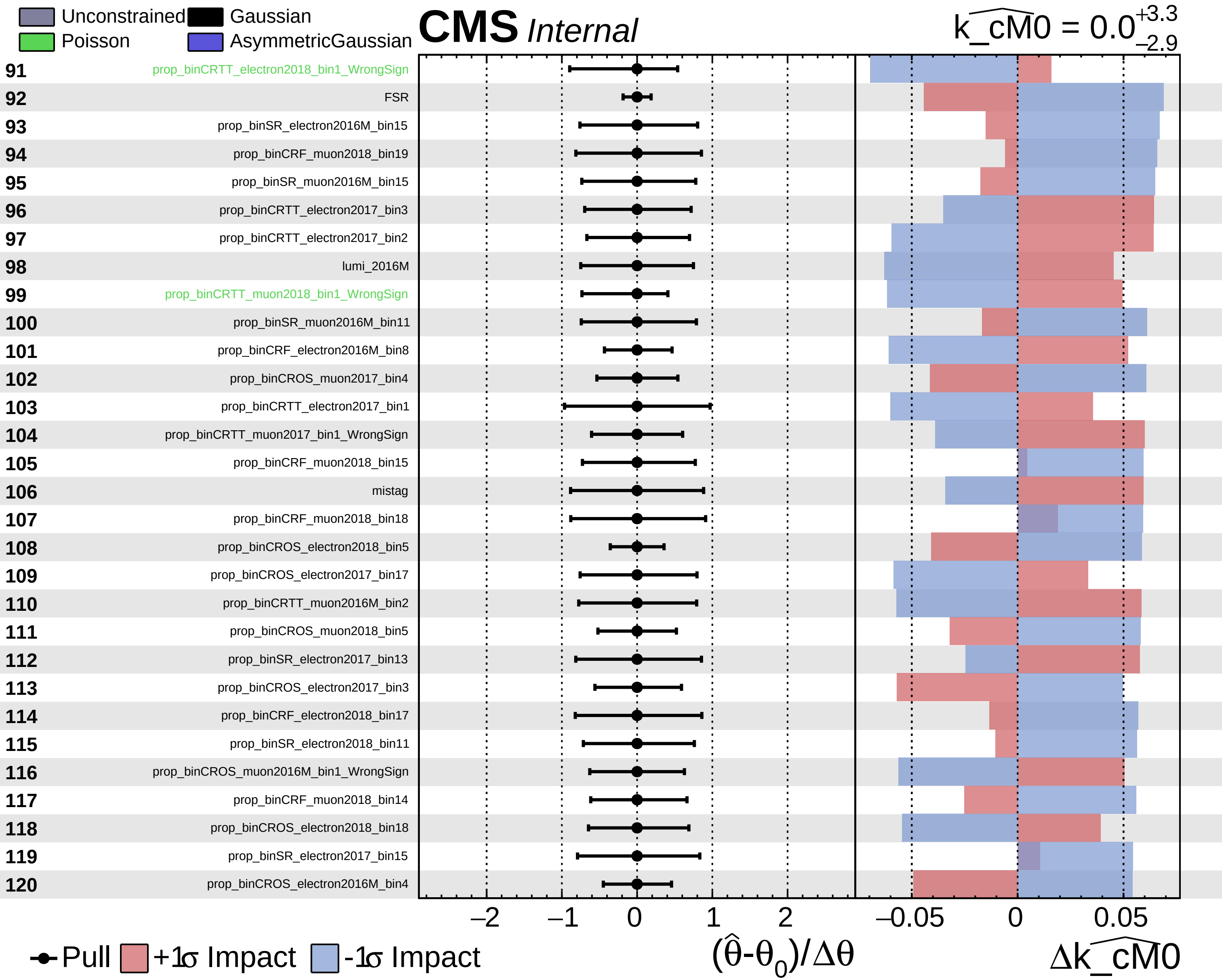

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
1 prop_binSR_muon2018_bin21
2 prop_binSR_electron2018_bin21
3 QCDScale_sig
4 prop_binSR_electron2017_bin21
5 prop_binSR_muon2017_bin21
6 pdf_Tot
7 OSest_muon_2017 $1.00^{+0.15}_{-0.15}$
8 PF
9 OSest_muon_2018 $1.00^{+0.11}_{-0.11}$
10 prop_binSR_muon2016M_bin21
11 OSest_electron_2018 $1.00^{+0.22}_{-0.22}$
12 OSest_electron_2016M $1.00^{+0.21}_{-0.21}$
13 OSest_electron_2017 $1.00^{+0.19}_{-0.19}$
14 OSest_muon_2016M $1.00^{+0.18}_{-0.18}$
15 prop_binSR_electron2016M_bin21
16 prop_binSR_electron2017_bin19
17 pu
18 prop_binSR_muon2018_bin20
19 prop_binSR_electron2017_bin17
20 prop_binSR_electron2018_bin20
21 prop_binCRF_muon2017_bin21_sm
22 prop_binSR_electron2016M_bin2_Fake
23 prop_binSR_muon2018_bin18
24 prop_binSR_electron2018_bin1_Fake
25 prop_binSR_electron2018_bin18
26 prop_binSR_muon2017_bin19
27 prop_binSR_electron2017_bin16
28 prop_binCROS_electron2018_bin1_WrongSign
29 prop_binSR_muon2017_bin17
30 prop_binSR_electron2017_bin18
−2 −1 0 1 2
−1 0 1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



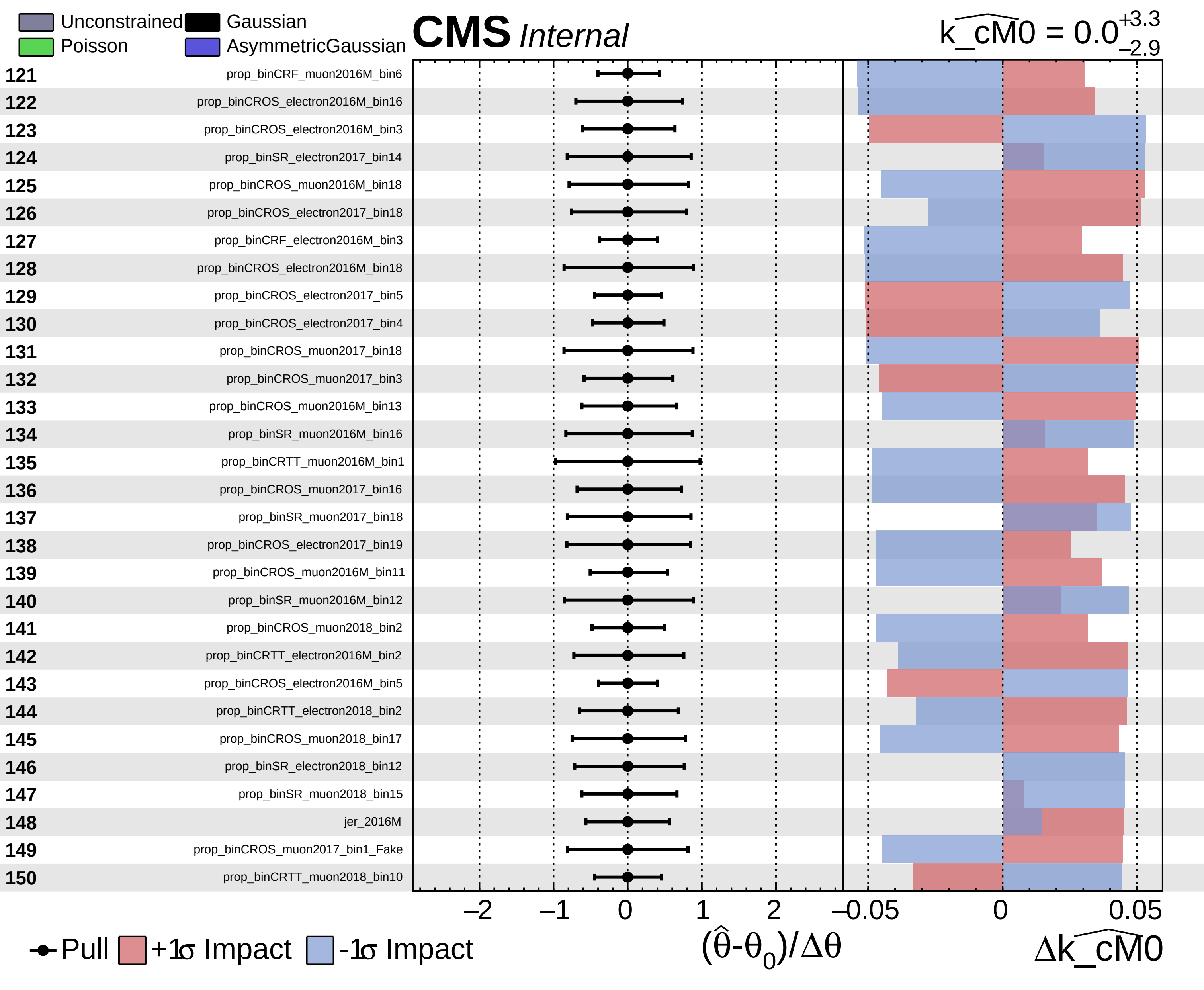
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
31 prop_binSR_electron2017_bin20
32 prop_binSR_muon2017_bin20
33 prop_binSR_muon2016M_bin19
34 prop_binSR_muon2018_bin19
35 prop_binSR_muon2018_bin2
36 prop_binSR_electron2017_bin4
37 prop_binSR_electron2016M_bin20
38 prop_binSR_electron2016M_bin3
39 prop_binSR_electron2018_bin2
40 prop_binSR_electron2018_bin19
41 prop_binSR_electron2016M_bin19
42 prop_binSR_muon2018_bin17
43 QCDScale_WrongSign
44 prop_binSR_electron2016M_bin18
45 prop_binSR_muon2018_bin1
46 prop_binSR_muon2017_bin2
47 prop_binSR_muon2017_bin16
48 prop_binSR_muon2017_bin15
49 prop_binSR_muon2016M_bin17
50 prop_binSR_muon2018_bin16
51 prop_binSR_electron2016M_bin5
52 prop_binSR_muon2017_bin3
53 prop_binCRF_muon2016M_bin1
54 prop_binCRF_electron2018_bin1
55 prop_binSR_electron2017_bin5
56 prop_binCRF_muon2017_bin1
57 prop_binSR_electron2016M_bin6
58 prop_binSR_muon2017_bin4
59 prop_binCRF_electron2016M_bin1
60 prop_binCRTT_electron2016M_bin1
−2 −1 0 1 2
−0.1 0 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



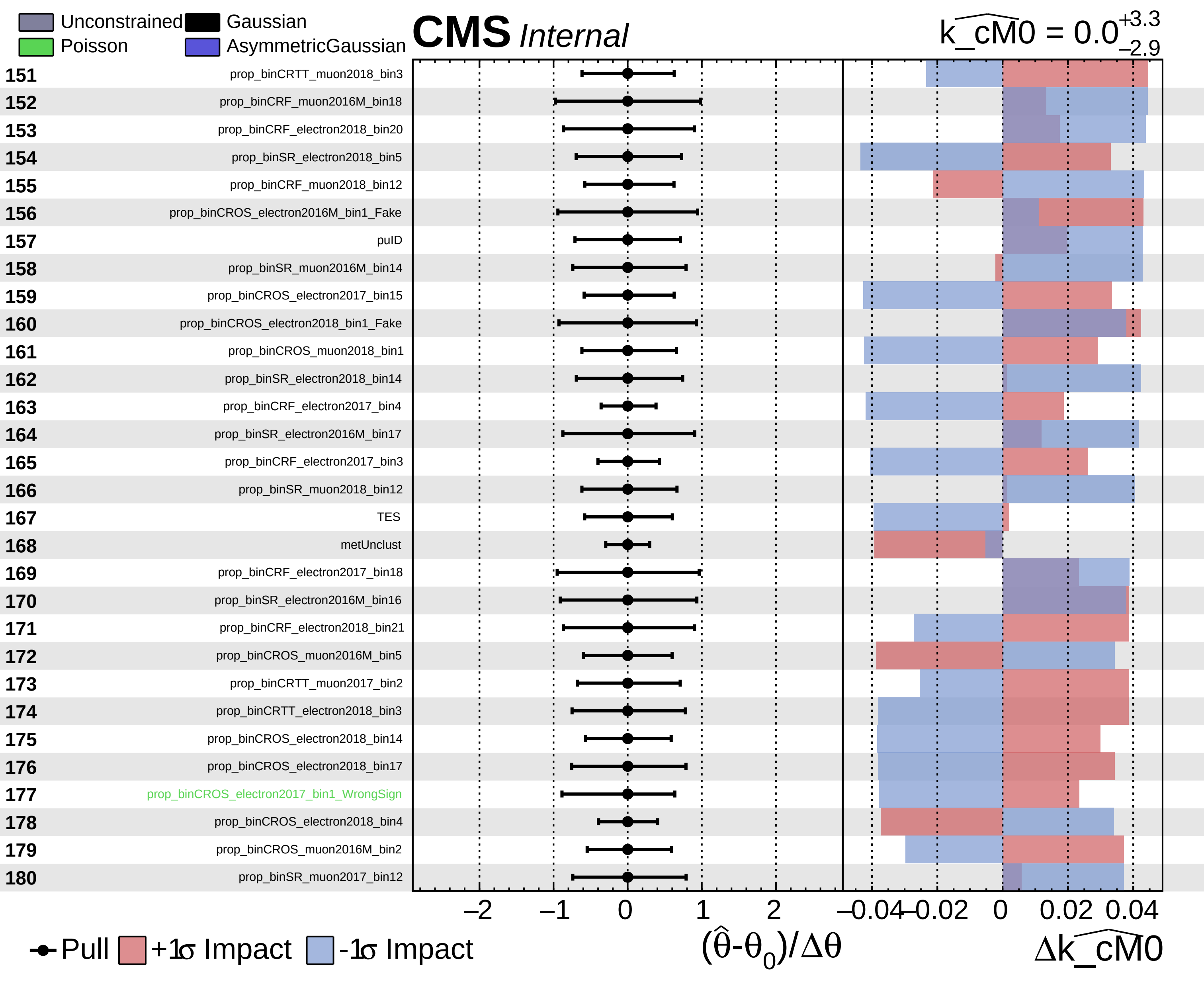
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
61 prop_binCRF_muon2017_bin20
62 prop_binCRF_electron2018_bin13
63 prop_binCROS_muon2018_bin11
64 prop_binSR_muon2016M_bin18
65 prop_binSR_electron2018_bin16
66 ISR
67 prop_binSR_muon2016M_bin7
68 prop_binCROS_electron2016M_bin1_WrongSign
69 prop_binCRF_muon2018_bin1
70 jer_2018
71 prop_binSR_muon2016M_bin20
72 prop_binSR_electron2018_bin3
73 prop_binCRF_muon2016M_bin12
74 prop_binCRF_electron2017_bin1
75 prop_binCRF_muon2018_bin21
76 prop_binSR_muon2018_bin4
77 prop_binSR_muon2017_bin6
78 prop_binSR_muon2018_bin13
79 prop_binSR_electron2018_bin10
80 prop_binSR_muon2018_bin7
81 prop_binSR_muon2018_bin6
82 prop_binCRF_muon2018_bin20
83 prop_binCROS_electron2016M_bin6
84 prop_binCROS_muon2017_bin17
85 prop_binSR_muon2018_bin14
86 prop_binCROS_muon2018_bin16
87 prop_binCRF_muon2016M_bin2
88 prop_binCROS_muon2018_bin18
89 prop_binSR_electron2018_bin17
90 prop_binSR_electron2018_bin15
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



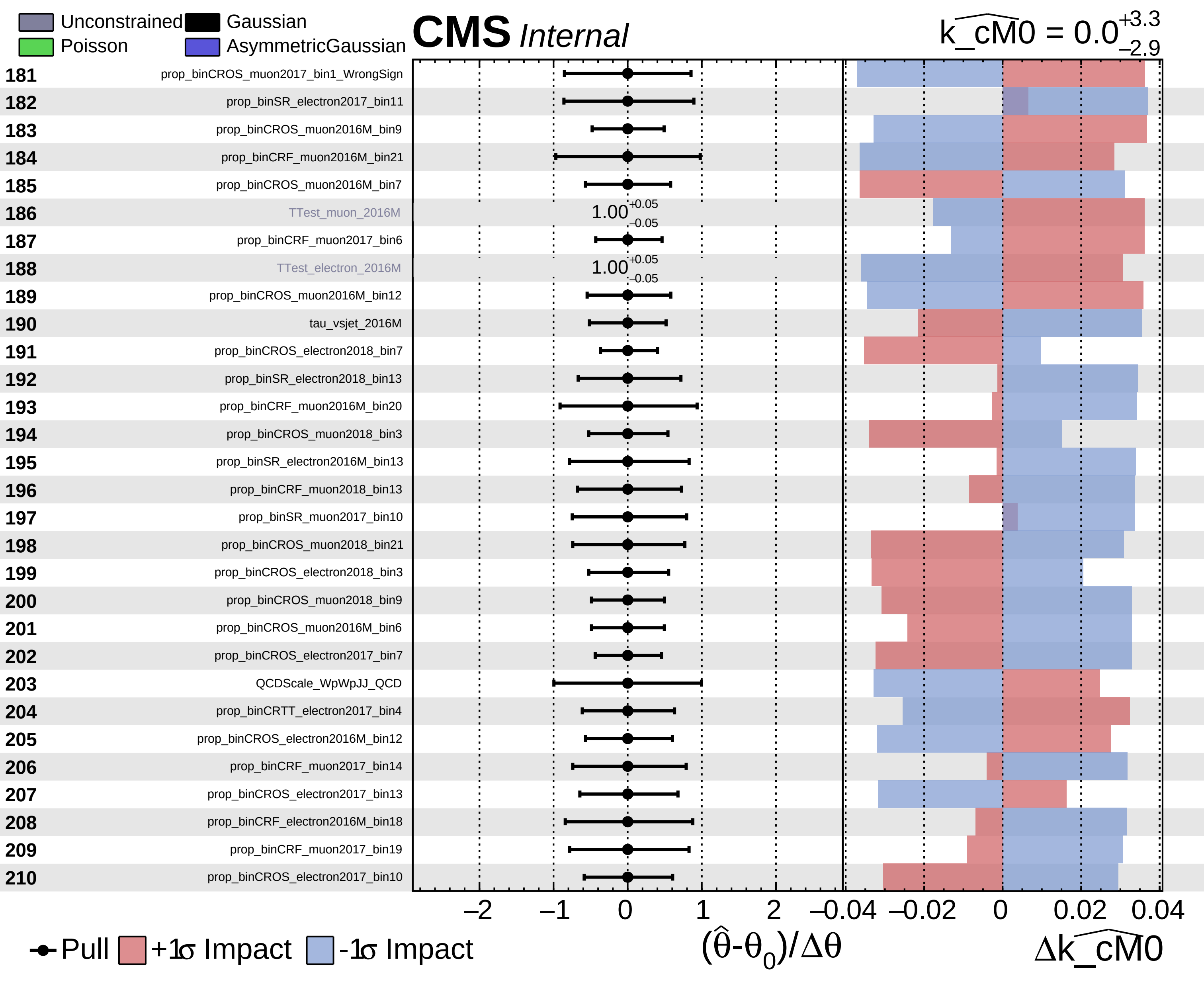
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
91 prop_binCRTT_electron2018_bin1_WrongSign
92 FSR
93 prop_binSR_electron2016M_bin15
94 prop_binCRF_muon2018_bin19
95 prop_binSR_muon2016M_bin15
96 prop_binCRTT_electron2017_bin3
97 prop_binCRTT_electron2017_bin2
98 lumi_2016M
99 prop_binCRTT_muon2018_bin1_WrongSign
100 prop_binSR_muon2016M_bin11
101 prop_binCRF_electron2016M_bin8
102 prop_binCROS_muon2017_bin4
103 prop_binCRTT_electron2017_bin1
104 prop_binCRTT_muon2017_bin1_WrongSign
105 prop_binCRF_muon2018_bin15
106 mistag
107 prop_binCRF_muon2018_bin18
108 prop_binCROS_electron2018_bin5
109 prop_binCROS_electron2017_bin17
110 prop_binCRTT_muon2016M_bin2
111 prop_binCROS_muon2018_bin5
112 prop_binSR_electron2017_bin13
113 prop_binCROS_electron2017_bin3
114 prop_binCRF_electron2018_bin17
115 prop_binSR_electron2018_bin11
116 prop_binCROS_muon2016M_bin1_WrongSign
117 prop_binCRF_muon2018_bin14
118 prop_binCROS_electron2018_bin18
119 prop_binSR_electron2017_bin15
120 prop_binCROS_electron2016M_bin4
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0



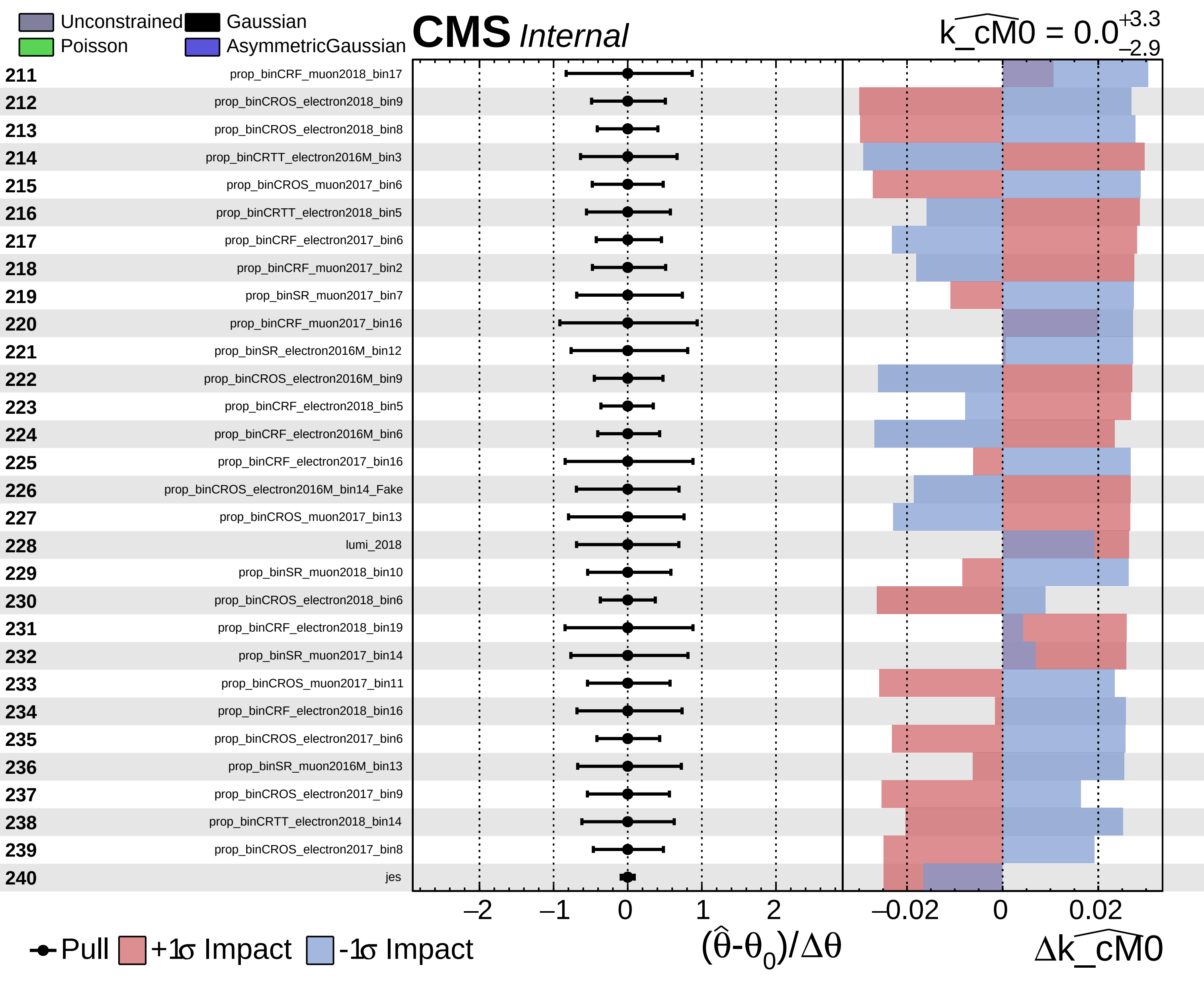
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
121 prop_binCRF_muon2016M_bin6
122 prop_binCROS_electron2016M_bin16
123 prop_binCROS_electron2016M_bin3
124 prop_binSR_electron2017_bin14
125 prop_binCROS_muon2016M_bin18
126 prop_binCROS_electron2017_bin18
127 prop_binCRF_electron2016M_bin3
128 prop_binCROS_electron2016M_bin18
129 prop_binCROS_electron2017_bin5
130 prop_binCROS_electron2017_bin4
131 prop_binCROS_muon2017_bin18
132 prop_binCROS_muon2017_bin3
133 prop_binCROS_muon2016M_bin13
134 prop_binSR_muon2016M_bin16
135 prop_binCRTT_muon2016M_bin1
136 prop_binCROS_muon2017_bin16
137 prop_binSR_muon2017_bin18
138 prop_binCROS_electron2017_bin19
139 prop_binCROS_muon2016M_bin11
140 prop_binSR_muon2016M_bin12
141 prop_binCROS_muon2018_bin2
142 prop_binCRTT_electron2016M_bin2
143 prop_binCROS_electron2016M_bin5
144 prop_binCRTT_electron2018_bin2
145 prop_binCROS_muon2018_bin17
146 prop_binSR_electron2018_bin12
147 prop_binSR_muon2018_bin15
148 jer_2016M
149 prop_binCROS_muon2017_bin1_Fake
150 prop_binCRTT_muon2018_bin10
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



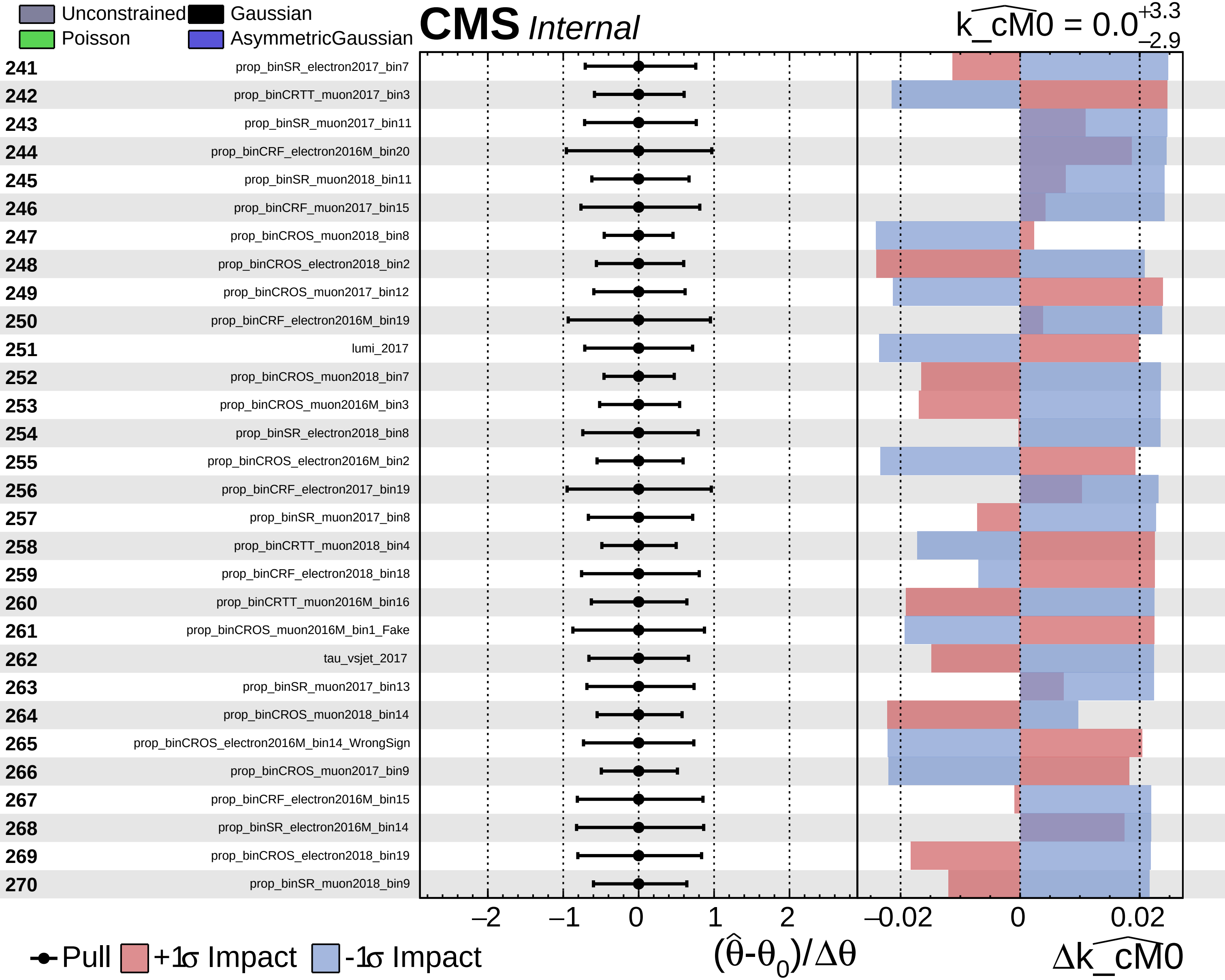
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
151 prop_binCRTT_muon2018_bin3
152 prop_binCRF_muon2016M_bin18
153 prop_binCRF_electron2018_bin20
154 prop_binSR_electron2018_bin5
155 prop_binCRF_muon2018_bin12
156 prop_binCROS_electron2016M_bin1_Fake
157 puID
158 prop_binSR_muon2016M_bin14
159 prop_binCROS_electron2017_bin15
160 prop_binCROS_electron2018_bin1_Fake
161 prop_binCROS_muon2018_bin1
162 prop_binSR_electron2018_bin14
163 prop_binCRF_electron2017_bin4
164 prop_binSR_electron2016M_bin17
165 prop_binCRF_electron2017_bin3
166 prop_binSR_muon2018_bin12
167 TES
168 metUnclust
169 prop_binCRF_electron2017_bin18
170 prop_binSR_electron2016M_bin16
171 prop_binCRF_electron2018_bin21
172 prop_binCROS_muon2016M_bin5
173 prop_binCRTT_muon2017_bin2
174 prop_binCRTT_electron2018_bin3
175 prop_binCROS_electron2018_bin14
176 prop_binCROS_electron2018_bin17
177 prop_binCROS_electron2017_bin1_WrongSign
178 prop_binCROS_electron2018_bin4
179 prop_binCROS_muon2016M_bin2
180 prop_binSR_muon2017_bin12
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0



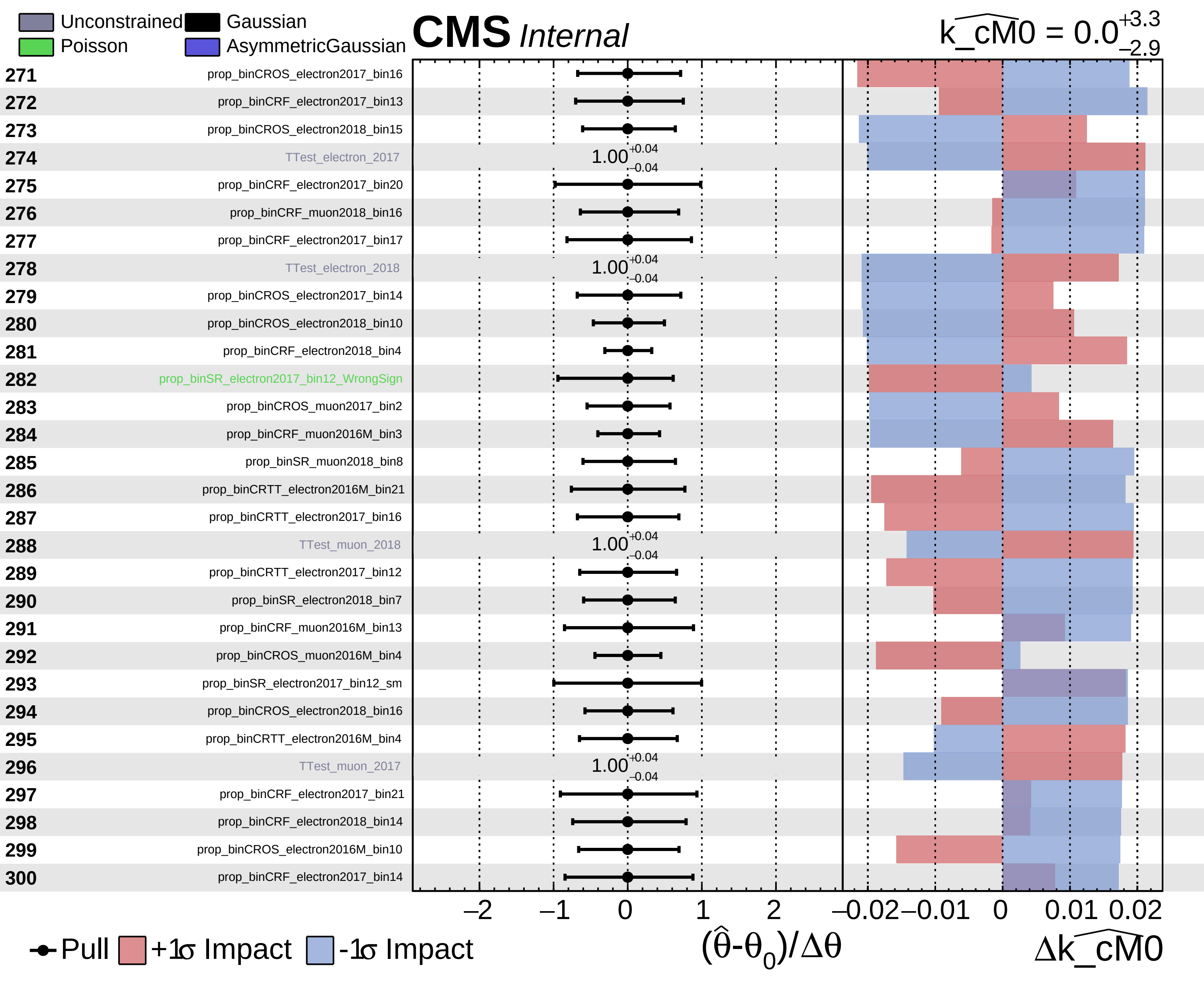
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
181 prop_binCROS_muon2017_bin1_WrongSign
182 prop_binSR_electron2017_bin11
183 prop_binCROS_muon2016M_bin9
184 prop_binCRF_muon2016M_bin21
185 prop_binCROS_muon2016M_bin7
186 TTest_muon_2016M
$1.00^{+0.05}_{-0.05}$
187 prop_binCRF_muon2017_bin6
188 TTest_electron_2016M
$1.00^{+0.05}_{-0.05}$
189 prop_binCROS_muon2016M_bin12
190 tau_vsjet_2016M
191 prop_binCROS_electron2018_bin7
192 prop_binSR_electron2018_bin13
193 prop_binCRF_muon2016M_bin20
194 prop_binCROS_muon2018_bin3
195 prop_binSR_electron2016M_bin13
196 prop_binCRF_muon2018_bin13
197 prop_binSR_muon2017_bin10
198 prop_binCROS_muon2018_bin21
199 prop_binCROS_electron2018_bin3
200 prop_binCROS_muon2018_bin9
201 prop_binCROS_muon2016M_bin6
202 prop_binCROS_electron2017_bin7
203 QCDScale_WpWpJJ_QCD
204 prop_binCRTT_electron2017_bin4
205 prop_binCROS_electron2016M_bin12
206 prop_binCRF_muon2017_bin14
207 prop_binCROS_electron2017_bin13
208 prop_binCRF_electron2016M_bin18
209 prop_binCRF_muon2017_bin19
210 prop_binCROS_electron2017_bin10
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

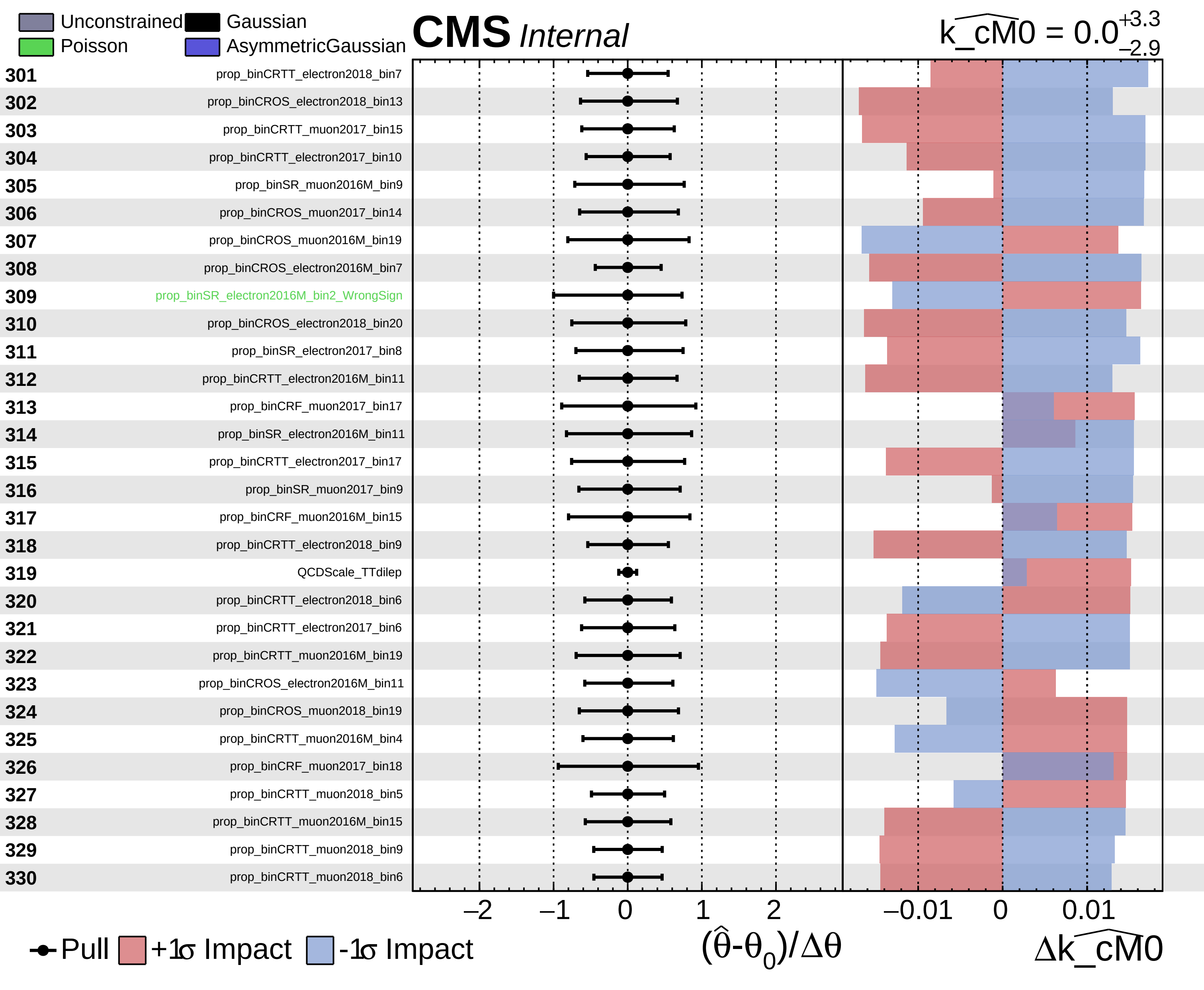


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
211 prop_binCRF_muon2018_bin17
212 prop_binCROS_electron2018_bin9
213 prop_binCROS_electron2018_bin8
214 prop_binCRTT_electron2016M_bin3
215 prop_binCROS_muon2017_bin6
216 prop_binCRTT_electron2018_bin5
217 prop_binCRF_electron2017_bin6
218 prop_binCRF_muon2017_bin2
219 prop_binSR_muon2017_bin7
220 prop_binCRF_muon2017_bin16
221 prop_binSR_electron2016M_bin12
222 prop_binCROS_electron2016M_bin9
223 prop_binCRF_electron2018_bin5
224 prop_binCRF_electron2016M_bin6
225 prop_binCRF_electron2017_bin16
226 prop_binCROS_electron2016M_bin14_Fake
227 prop_binCROS_muon2017_bin13
228 lumi_2018
229 prop_binSR_muon2018_bin10
230 prop_binCROS_electron2018_bin6
231 prop_binCRF_electron2018_bin19
232 prop_binSR_muon2017_bin14
233 prop_binCROS_muon2017_bin11
234 prop_binCRF_electron2018_bin16
235 prop_binCROS_electron2017_bin6
236 prop_binSR_muon2016M_bin13
237 prop_binCROS_electron2017_bin9
238 prop_binCRTT_electron2018_bin14
239 prop_binCROS_electron2017_bin8
240 jes
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM0

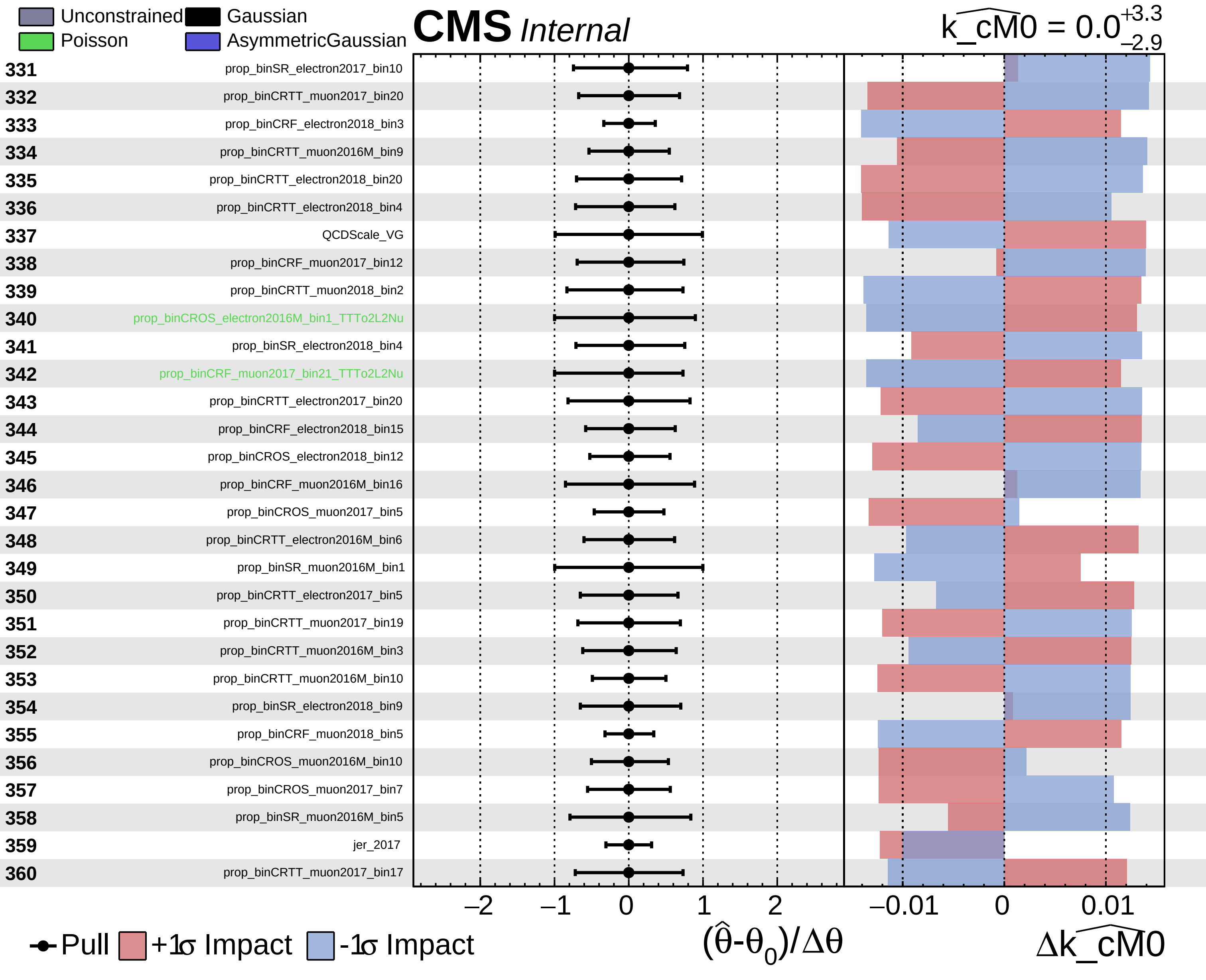


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
241 prop_binSR_electron2017_bin7
242 prop_binCRTT_muon2017_bin3
243 prop_binSR_muon2017_bin11
244 prop_binCRF_electron2016M_bin20
245 prop_binSR_muon2018_bin11
246 prop_binCRF_muon2017_bin15
247 prop_binCROS_muon2018_bin8
248 prop_binCROS_electron2018_bin2
249 prop_binCROS_muon2017_bin12
250 prop_binCRF_electron2016M_bin19
251 lumi_2017
252 prop_binCROS_muon2018_bin7
253 prop_binCROS_muon2016M_bin3
254 prop_binSR_electron2018_bin8
255 prop_binCROS_electron2016M_bin2
256 prop_binCRF_electron2017_bin19
257 prop_binSR_muon2017_bin8
258 prop_binCRTT_muon2018_bin4
259 prop_binCRF_electron2018_bin18
260 prop_binCRTT_muon2016M_bin16
261 prop_binCROS_muon2016M_bin1_Fake
262 tau_vsjet_2017
263 prop_binSR_muon2017_bin13
264 prop_binCROS_muon2018_bin14
265 prop_binCROS_electron2016M_bin14_WrongSign
266 prop_binCROS_muon2017_bin9
267 prop_binCRF_electron2016M_bin15
268 prop_binSR_electron2016M_bin14
269 prop_binCROS_electron2018_bin19
270 prop_binSR_muon2018_bin9
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

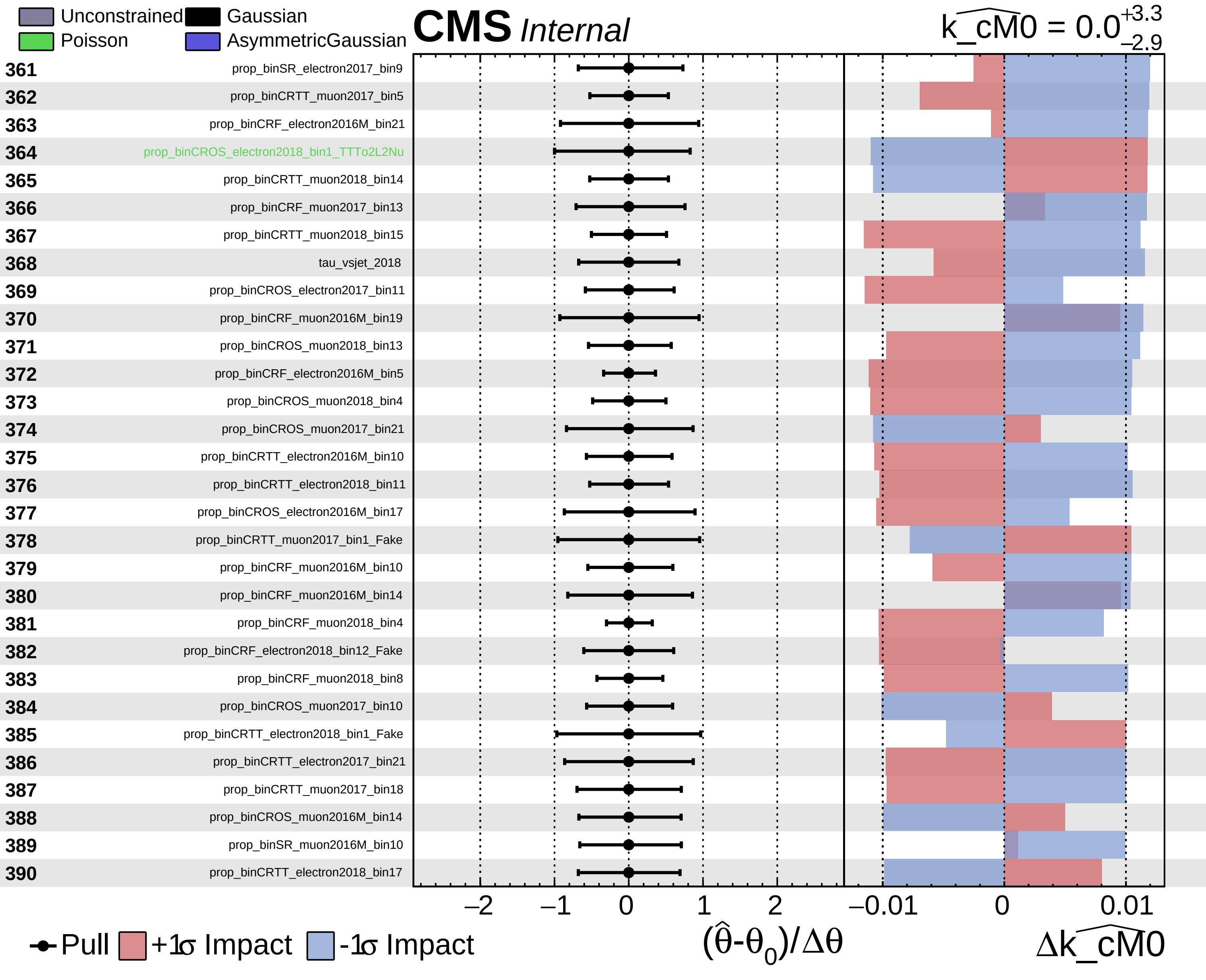

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
271 prop_binCROS_electron2017_bin16
272 prop_binCRF_electron2017_bin13
273 prop_binCROS_electron2018_bin15
274 TTest_electron_2017 1.00 +0.04 −0.04
275 prop_binCRF_electron2017_bin20
276 prop_binCRF_muon2018_bin16
277 prop_binCRF_electron2017_bin17
278 TTest_electron_2018 1.00 +0.04 −0.04
279 prop_binCROS_electron2017_bin14
280 prop_binCROS_electron2018_bin10
281 prop_binCRF_electron2018_bin4
282 prop_binSR_electron2017_bin12_WrongSign
283 prop_binCROS_muon2017_bin2
284 prop_binCRF_muon2016M_bin3
285 prop_binSR_muon2018_bin8
286 prop_binCRTT_electron2016M_bin21
287 prop_binCRTT_electron2017_bin16
288 TTest_muon_2018 1.00 +0.04 −0.04
289 prop_binCRTT_electron2017_bin12
290 prop_binSR_electron2018_bin7
291 prop_binCRF_muon2016M_bin13
292 prop_binCROS_muon2016M_bin4
293 prop_binSR_electron2017_bin12_sm
294 prop_binCROS_electron2018_bin16
295 prop_binCRTT_electron2016M_bin4
296 TTest_muon_2017 1.00 +0.04 −0.04
297 prop_binCRF_electron2017_bin21
298 prop_binCRF_electron2018_bin14
299 prop_binCROS_electron2016M_bin10
300 prop_binCRF_electron2017_bin14
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



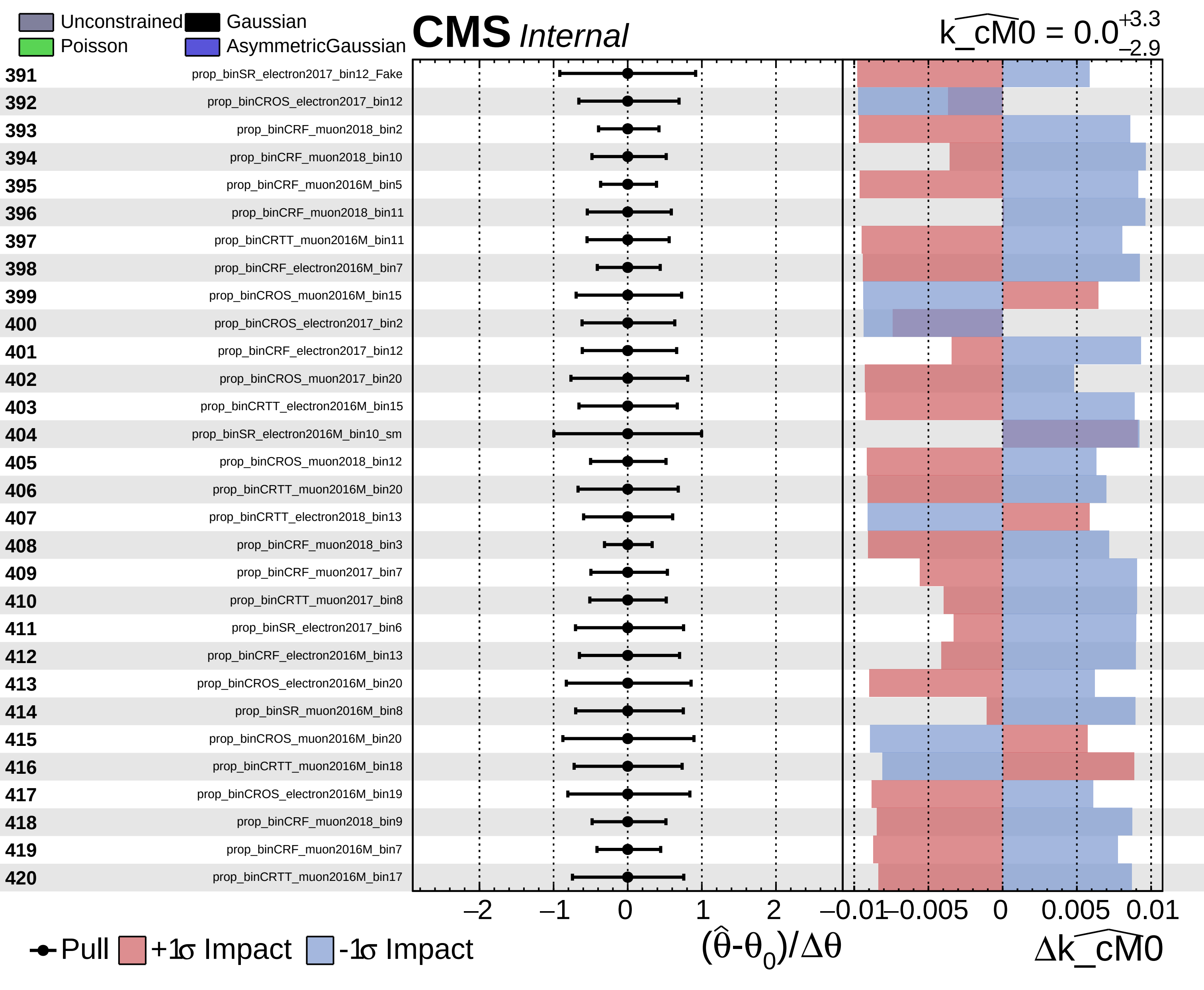
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
301 prop_binCRTT_electron2018_bin7
302 prop_binCROS_electron2018_bin13
303 prop_binCRTT_muon2017_bin15
304 prop_binCRTT_electron2017_bin10
305 prop_binSR_muon2016M_bin9
306 prop_binCROS_muon2017_bin14
307 prop_binCROS_muon2016M_bin19
308 prop_binCROS_electron2016M_bin7
309 prop_binSR_electron2016M_bin2_WrongSign
310 prop_binCROS_electron2018_bin20
311 prop_binSR_electron2017_bin8
312 prop_binCRTT_electron2016M_bin11
313 prop_binCRF_muon2017_bin17
314 prop_binSR_electron2016M_bin11
315 prop_binCRTT_electron2017_bin17
316 prop_binSR_muon2017_bin9
317 prop_binCRF_muon2016M_bin15
318 prop_binCRTT_electron2018_bin9
319 QCDScale_TTdilep
320 prop_binCRTT_electron2018_bin6
321 prop_binCRTT_electron2017_bin6
322 prop_binCRTT_muon2016M_bin19
323 prop_binCROS_electron2016M_bin11
324 prop_binCROS_muon2018_bin19
325 prop_binCRTT_muon2016M_bin4
326 prop_binCRF_muon2017_bin18
327 prop_binCRTT_muon2018_bin5
328 prop_binCRTT_muon2016M_bin15
329 prop_binCRTT_muon2018_bin9
330 prop_binCRTT_muon2018_bin6
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



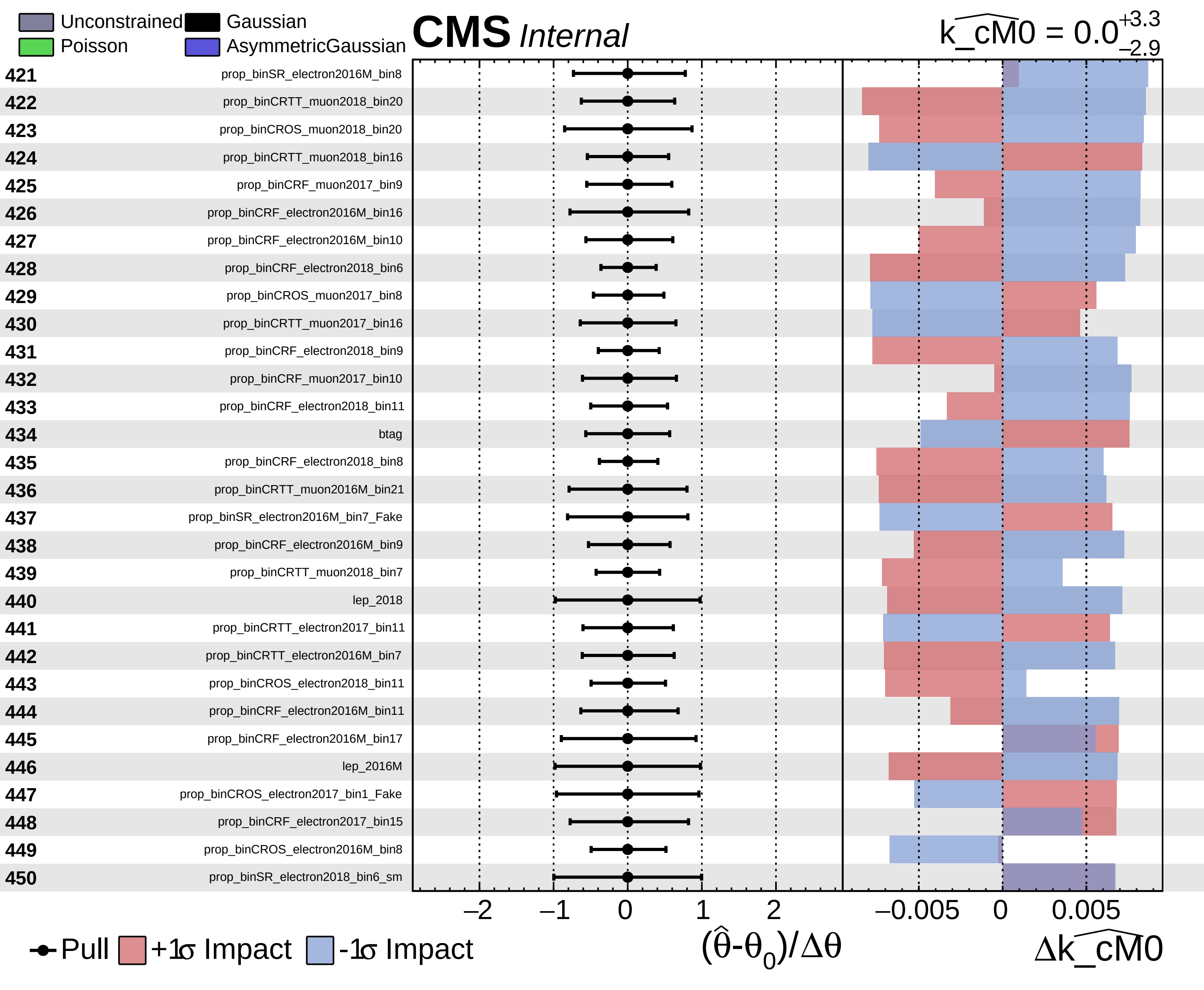
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
331 prop_binSR_electron2017_bin10
332 prop_binCRTT_muon2017_bin20
333 prop_binCRF_electron2018_bin3
334 prop_binCRTT_muon2016M_bin9
335 prop_binCRTT_electron2018_bin20
336 prop_binCRTT_electron2018_bin4
337 QCDScale_VG
338 prop_binCRF_muon2017_bin12
339 prop_binCRTT_muon2018_bin2
340 prop_binCROS_electron2016M_bin1_TTTo2L2Nu
341 prop_binSR_electron2018_bin4
342 prop_binCRF_muon2017_bin21_TTTo2L2Nu
343 prop_binCRTT_electron2017_bin20
344 prop_binCRF_electron2018_bin15
345 prop_binCROS_electron2018_bin12
346 prop_binCRF_muon2016M_bin16
347 prop_binCROS_muon2017_bin5
348 prop_binCRTT_electron2016M_bin6
349 prop_binSR_muon2016M_bin1
350 prop_binCRTT_electron2017_bin5
351 prop_binCRTT_muon2017_bin19
352 prop_binCRTT_muon2016M_bin3
353 prop_binCRTT_muon2016M_bin10
354 prop_binSR_electron2018_bin9
355 prop_binCRF_muon2018_bin5
356 prop_binCROS_muon2016M_bin10
357 prop_binCROS_muon2017_bin7
358 prop_binSR_muon2016M_bin5
359 jer_2017
360 prop_binCRTT_muon2017_bin17
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0



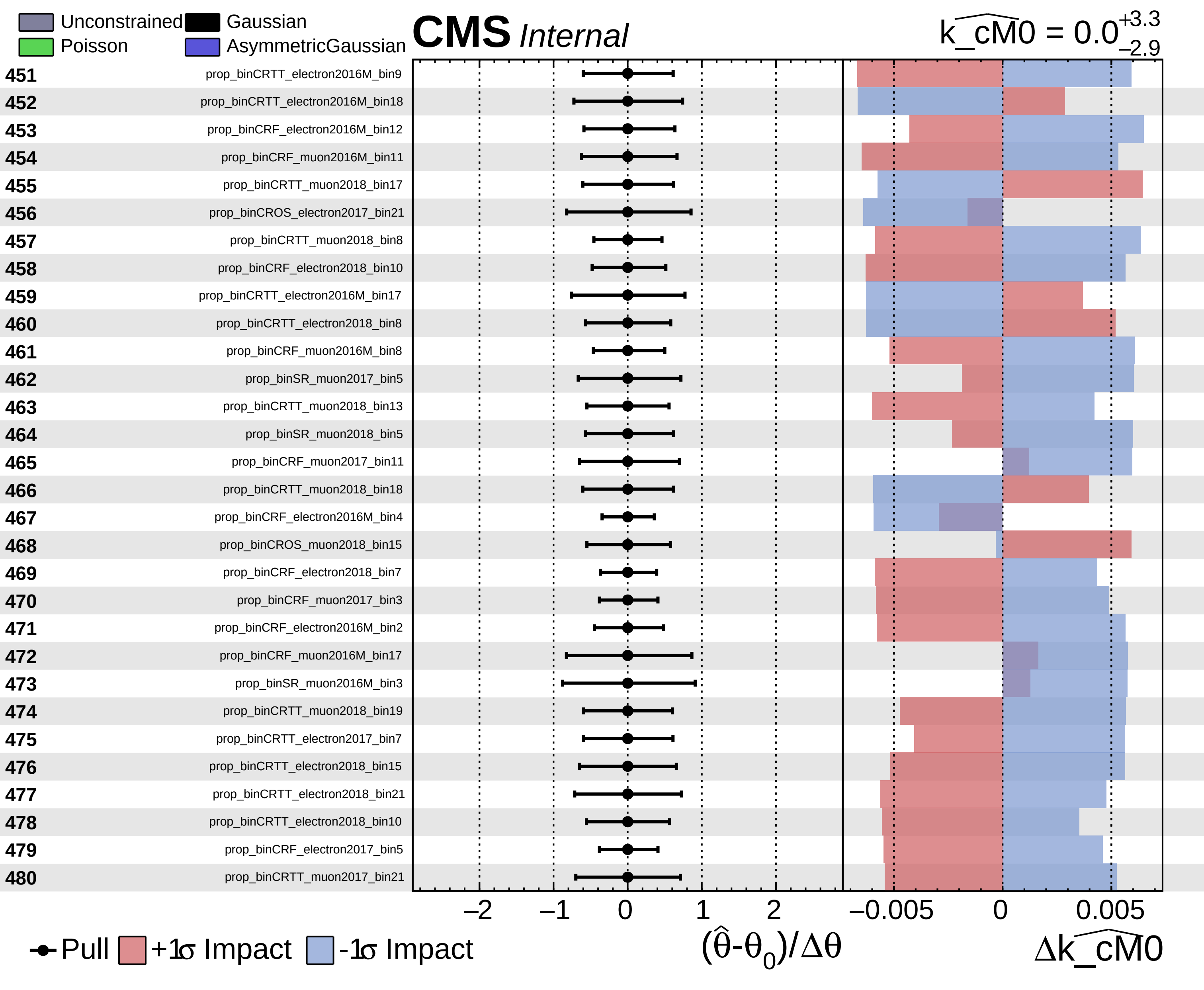
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
361 prop_binSR_electron2017_bin9
362 prop_binCRTT_muon2017_bin5
363 prop_binCRF_electron2016M_bin21
364 prop_binCROS_electron2018_bin1_TTTo2L2Nu
365 prop_binCRTT_muon2018_bin14
366 prop_binCRF_muon2017_bin13
367 prop_binCRTT_muon2018_bin15
368 tau_vsjet_2018
369 prop_binCROS_electron2017_bin11
370 prop_binCRF_muon2016M_bin19
371 prop_binCROS_muon2018_bin13
372 prop_binCRF_electron2016M_bin5
373 prop_binCROS_muon2018_bin4
374 prop_binCROS_muon2017_bin21
375 prop_binCRTT_electron2016M_bin10
376 prop_binCRTT_electron2018_bin11
377 prop_binCROS_electron2016M_bin17
378 prop_binCRTT_muon2017_bin1_Fake
379 prop_binCRF_muon2016M_bin10
380 prop_binCRF_muon2016M_bin14
381 prop_binCRF_muon2018_bin4
382 prop_binCRF_electron2018_bin12_Fake
383 prop_binCRF_muon2018_bin8
384 prop_binCROS_muon2017_bin10
385 prop_binCRTT_electron2018_bin1_Fake
386 prop_binCRTT_electron2017_bin21
387 prop_binCRTT_muon2017_bin18
388 prop_binCROS_muon2016M_bin14
389 prop_binSR_muon2016M_bin10
390 prop_binCRTT_electron2018_bin17
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

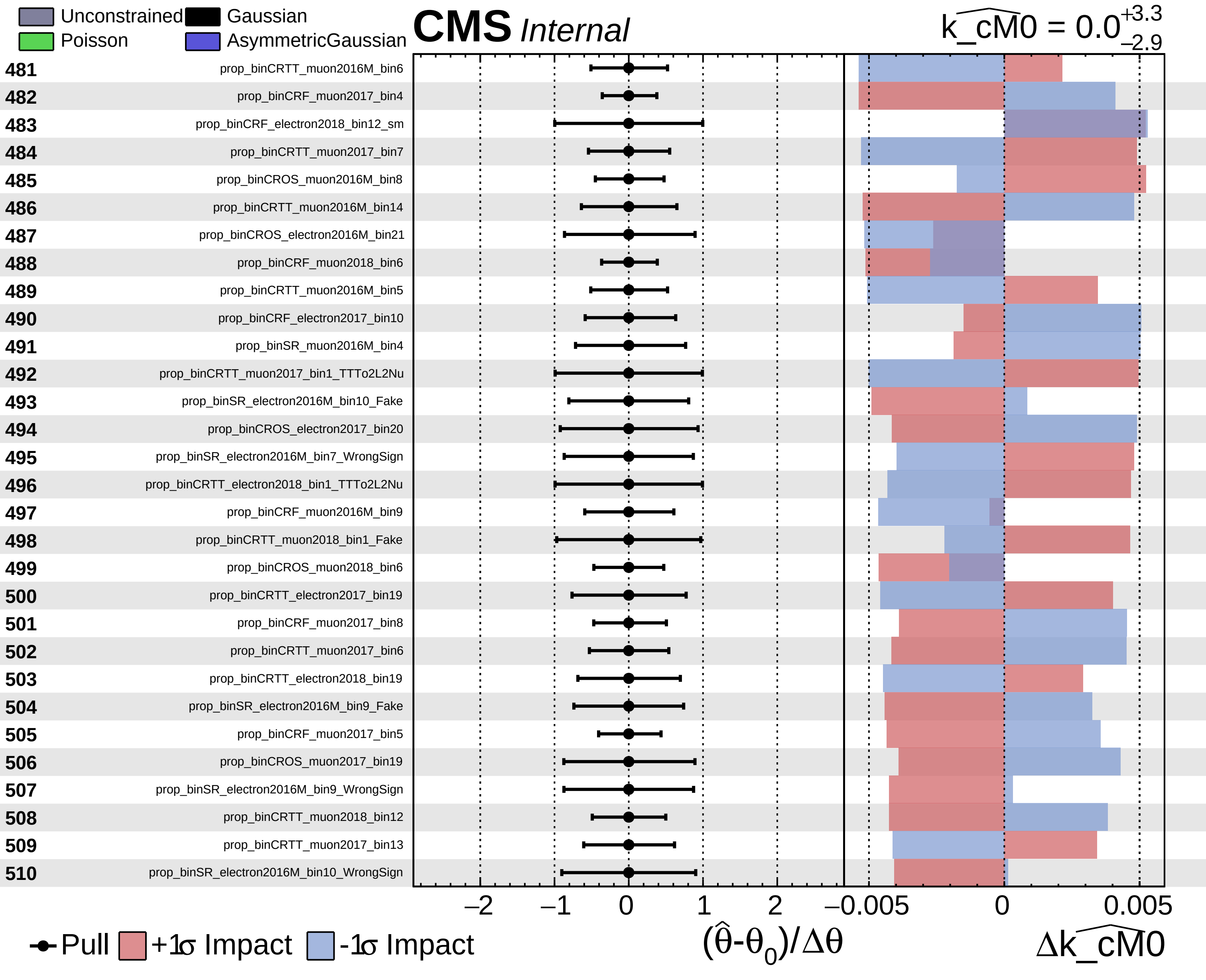


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
391 prop_binSR_electron2017_bin12_Fake
392 prop_binCROS_electron2017_bin12
393 prop_binCRF_muon2018_bin2
394 prop_binCRF_muon2018_bin10
395 prop_binCRF_muon2016M_bin5
396 prop_binCRF_muon2018_bin11
397 prop_binCRTT_muon2016M_bin11
398 prop_binCRF_electron2016M_bin7
399 prop_binCROS_muon2016M_bin15
400 prop_binCROS_electron2017_bin2
401 prop_binCRF_electron2017_bin12
402 prop_binCROS_muon2017_bin20
403 prop_binCRTT_electron2016M_bin15
404 prop_binSR_electron2016M_bin10_sm
405 prop_binCROS_muon2018_bin12
406 prop_binCRTT_muon2016M_bin20
407 prop_binCRTT_electron2018_bin13
408 prop_binCRF_muon2018_bin3
409 prop_binCRF_muon2017_bin7
410 prop_binCRTT_muon2017_bin8
411 prop_binSR_electron2017_bin6
412 prop_binCRF_electron2016M_bin13
413 prop_binCROS_electron2016M_bin20
414 prop_binSR_muon2016M_bin8
415 prop_binCROS_muon2016M_bin20
416 prop_binCRTT_muon2016M_bin18
417 prop_binCROS_electron2016M_bin19
418 prop_binCRF_muon2018_bin9
419 prop_binCRF_muon2016M_bin7
420 prop_binCRTT_muon2016M_bin17
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

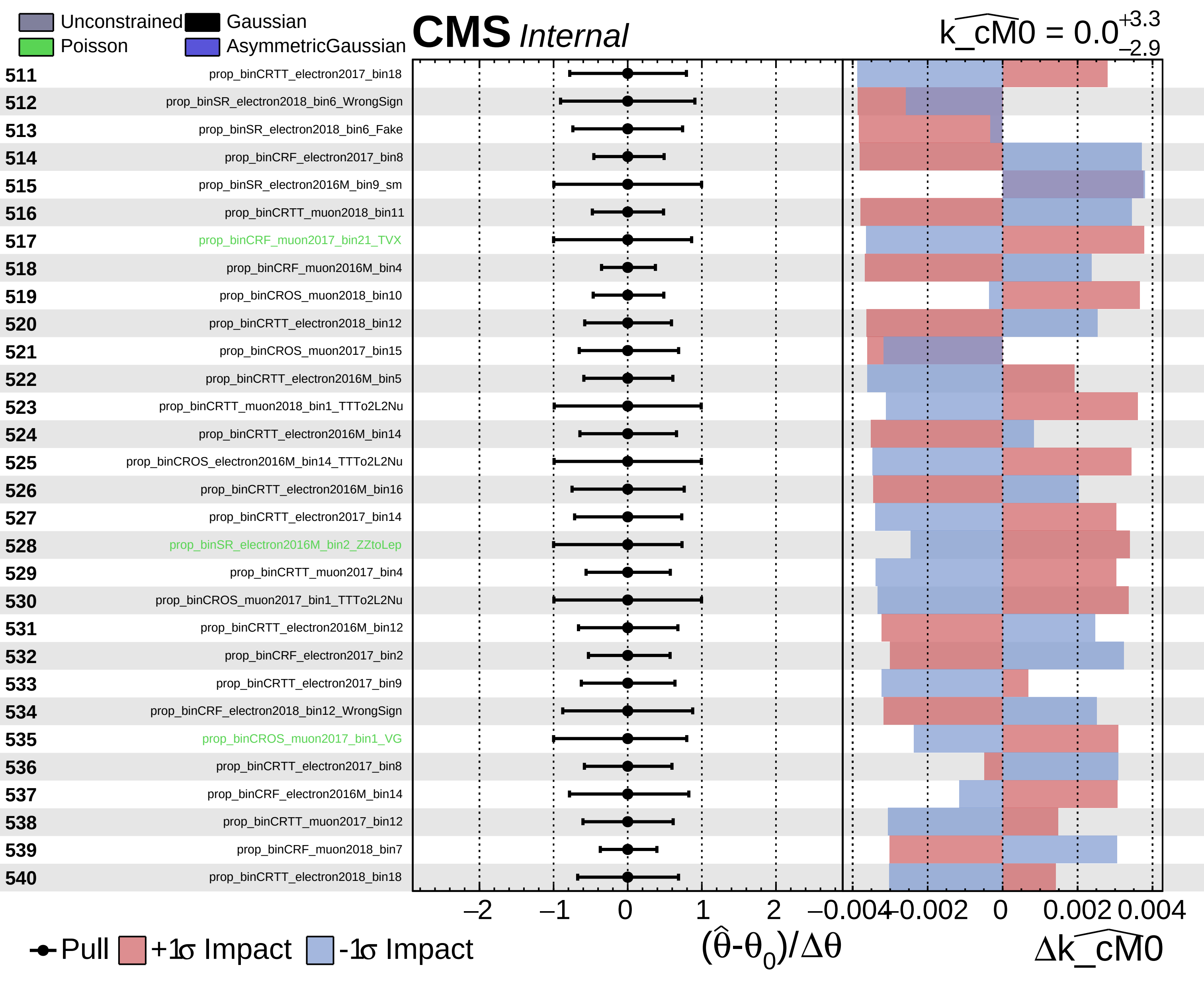


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
421 prop_binSR_electron2016M_bin8
422 prop_binCRTT_muon2018_bin20
423 prop_binCROS_muon2018_bin20
424 prop_binCRTT_muon2018_bin16
425 prop_binCRF_muon2017_bin9
426 prop_binCRF_electron2016M_bin16
427 prop_binCRF_electron2016M_bin10
428 prop_binCRF_electron2018_bin6
429 prop_binCROS_muon2017_bin8
430 prop_binCRTT_muon2017_bin16
431 prop_binCRF_electron2018_bin9
432 prop_binCRF_muon2017_bin10
433 prop_binCRF_electron2018_bin11
434 btag
435 prop_binCRF_electron2018_bin8
436 prop_binCRTT_muon2016M_bin21
437 prop_binSR_electron2016M_bin7_Fake
438 prop_binCRF_electron2016M_bin9
439 prop_binCRTT_muon2018_bin7
440 lep_2018
441 prop_binCRTT_electron2017_bin11
442 prop_binCRTT_electron2016M_bin7
443 prop_binCROS_electron2018_bin11
444 prop_binCRF_electron2016M_bin11
445 prop_binCRF_electron2016M_bin17
446 lep_2016M
447 prop_binCROS_electron2017_bin1_Fake
448 prop_binCRF_electron2017_bin15
449 prop_binCROS_electron2016M_bin8
450 prop_binSR_electron2018_bin6_sm
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

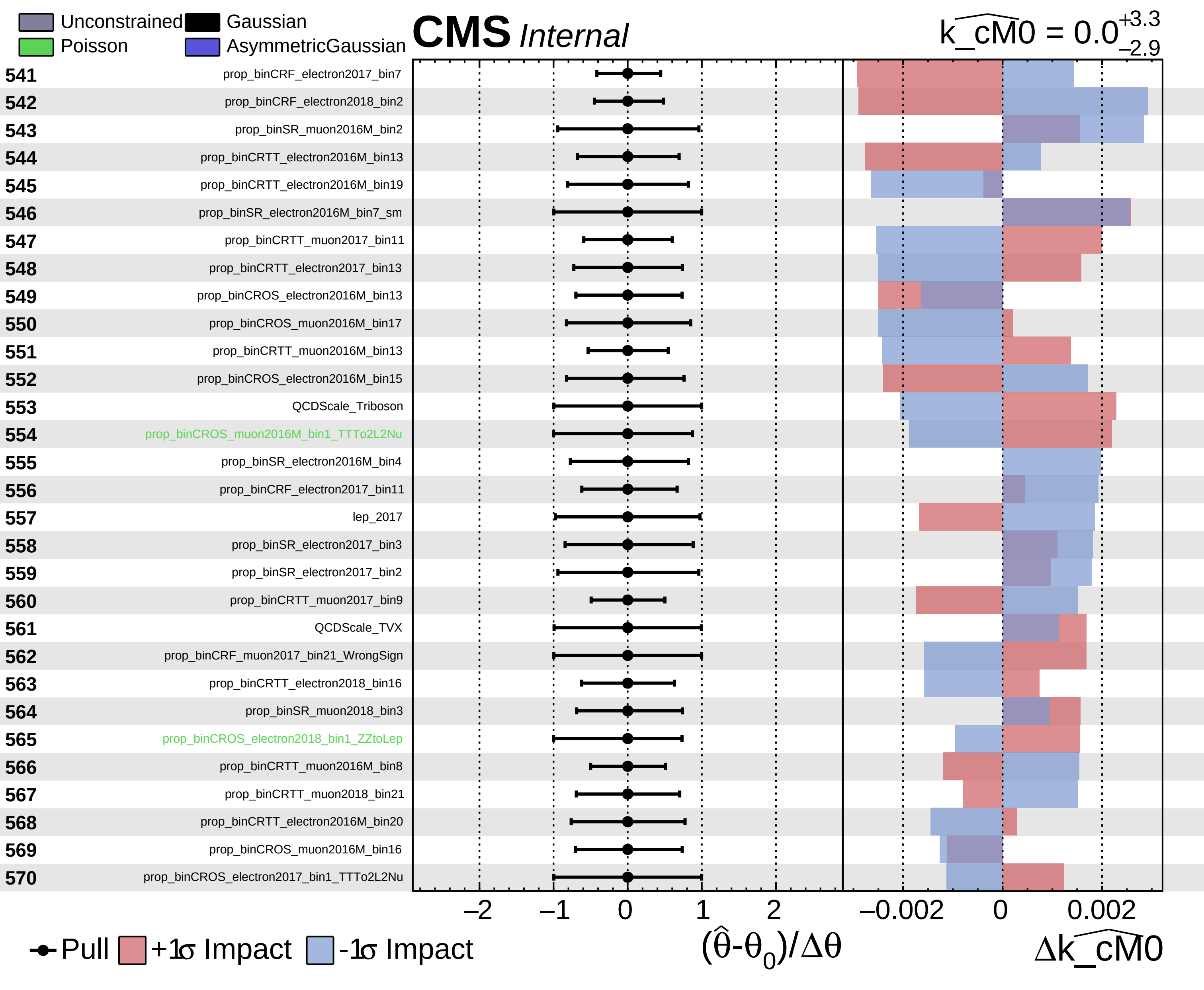

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
451 prop_binCRTT_electron2016M_bin9
452 prop_binCRTT_electron2016M_bin18
453 prop_binCRF_electron2016M_bin12
454 prop_binCRF_muon2016M_bin11
455 prop_binCRTT_muon2018_bin17
456 prop_binCROS_electron2017_bin21
457 prop_binCRTT_muon2018_bin8
458 prop_binCRF_electron2018_bin10
459 prop_binCRTT_electron2016M_bin17
460 prop_binCRTT_electron2018_bin8
461 prop_binCRF_muon2016M_bin8
462 prop_binSR_muon2017_bin5
463 prop_binCRTT_muon2018_bin13
464 prop_binSR_muon2018_bin5
465 prop_binCRF_muon2017_bin11
466 prop_binCRTT_muon2018_bin18
467 prop_binCRF_electron2016M_bin4
468 prop_binCROS_muon2018_bin15
469 prop_binCRF_electron2018_bin7
470 prop_binCRF_muon2017_bin3
471 prop_binCRF_electron2016M_bin2
472 prop_binCRF_muon2016M_bin17
473 prop_binSR_muon2016M_bin3
474 prop_binCRTT_muon2018_bin19
475 prop_binCRTT_electron2017_bin7
476 prop_binCRTT_electron2018_bin15
477 prop_binCRTT_electron2018_bin21
478 prop_binCRTT_electron2018_bin10
479 prop_binCRF_electron2017_bin5
480 prop_binCRTT_muon2017_bin21
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



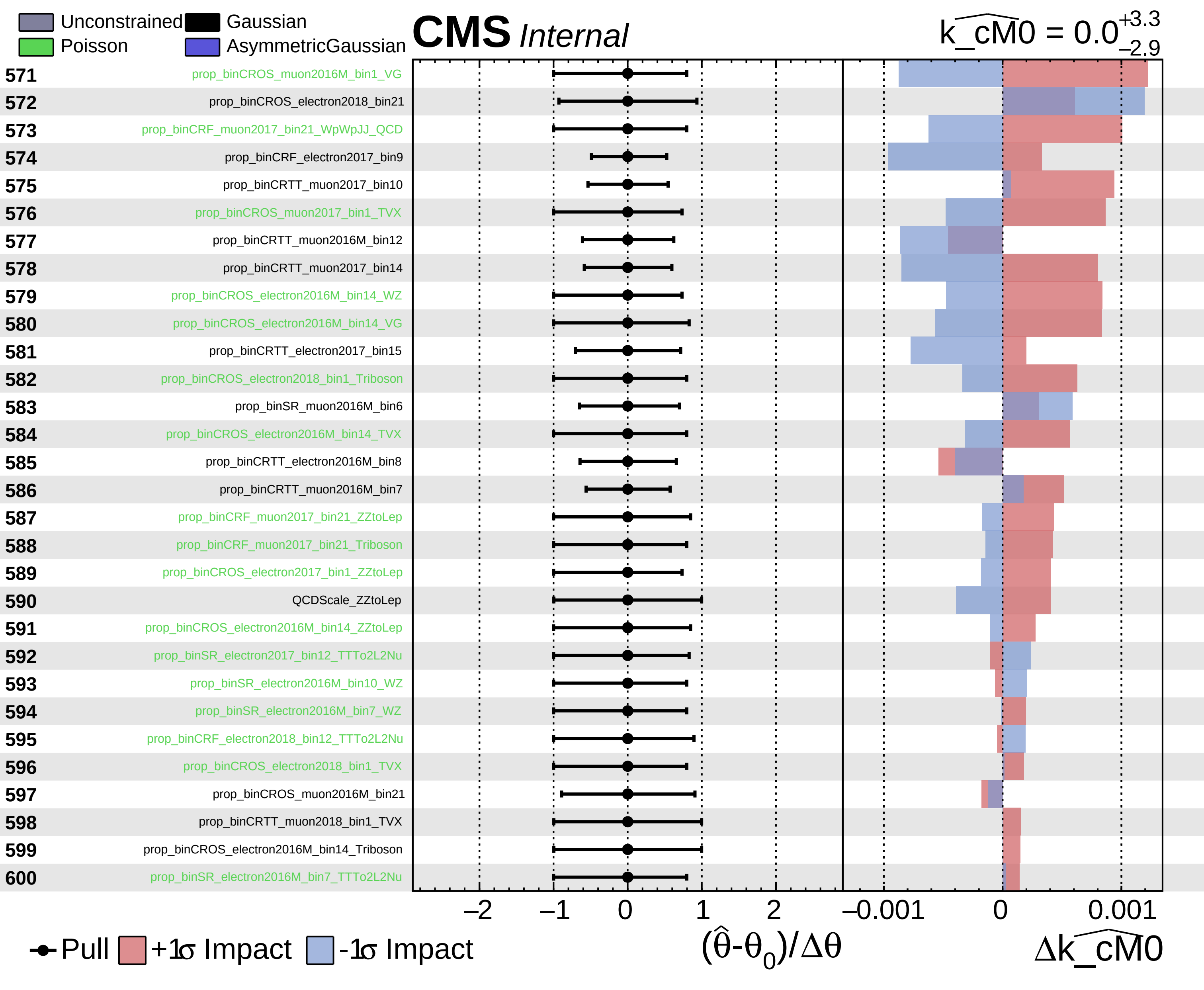
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
481 prop_binCRTT_muon2016M_bin6
482 prop_binCRF_muon2017_bin4
483 prop_binCRF_electron2018_bin12_sm
484 prop_binCRTT_muon2017_bin7
485 prop_binCROS_muon2016M_bin8
486 prop_binCRTT_muon2016M_bin14
487 prop_binCROS_electron2016M_bin21
488 prop_binCRF_muon2018_bin6
489 prop_binCRTT_muon2016M_bin5
490 prop_binCRF_electron2017_bin10
491 prop_binSR_muon2016M_bin4
492 prop_binCRTT_muon2017_bin1_TTTo2L2Nu
493 prop_binSR_electron2016M_bin10_Fake
494 prop_binCROS_electron2017_bin20
495 prop_binSR_electron2016M_bin7_WrongSign
496 prop_binCRTT_electron2018_bin1_TTTo2L2Nu
497 prop_binCRF_muon2016M_bin9
498 prop_binCRTT_muon2018_bin1_Fake
499 prop_binCROS_muon2018_bin6
500 prop_binCRTT_electron2017_bin19
501 prop_binCRF_muon2017_bin8
502 prop_binCRTT_muon2017_bin6
503 prop_binCRTT_electron2018_bin19
504 prop_binSR_electron2016M_bin9_Fake
505 prop_binCRF_muon2017_bin5
506 prop_binCROS_muon2017_bin19
507 prop_binSR_electron2016M_bin9_WrongSign
508 prop_binCRTT_muon2018_bin12
509 prop_binCRTT_muon2017_bin13
510 prop_binSR_electron2016M_bin10_WrongSign
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

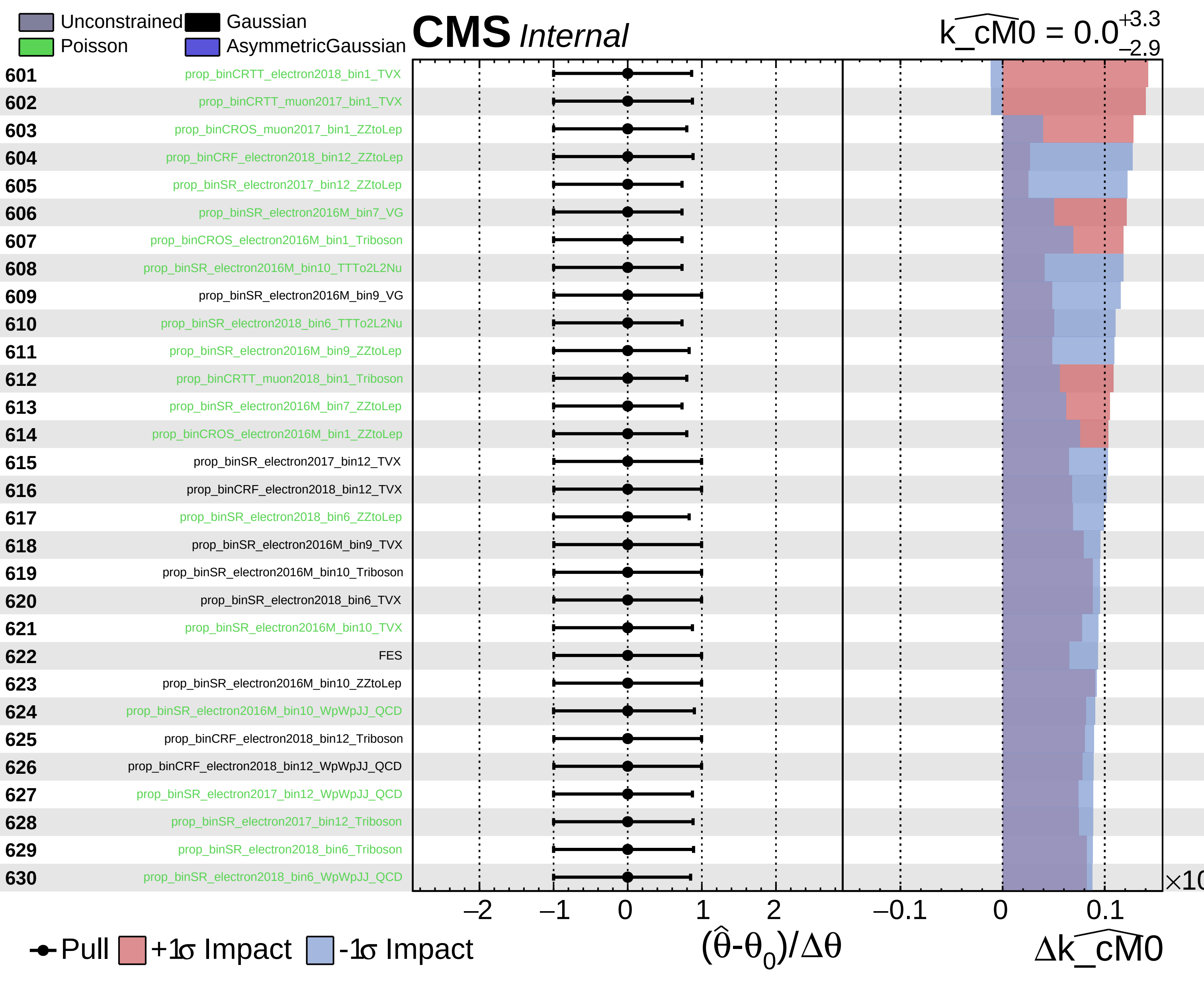


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
511 prop_binCRTT_electron2017_bin18
512 prop_binSR_electron2018_bin6_WrongSign
513 prop_binSR_electron2018_bin6_Fake
514 prop_binCRF_electron2017_bin8
515 prop_binSR_electron2016M_bin9_sm
516 prop_binCRTT_muon2018_bin11
517 prop_binCRF_muon2017_bin21_TVX
518 prop_binCRF_muon2016M_bin4
519 prop_binCROS_muon2018_bin10
520 prop_binCRTT_electron2018_bin12
521 prop_binCROS_muon2017_bin15
522 prop_binCRTT_electron2016M_bin5
523 prop_binCRTT_muon2018_bin1_TTTo2L2Nu
524 prop_binCRTT_electron2016M_bin14
525 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
526 prop_binCRTT_electron2016M_bin16
527 prop_binCRTT_electron2017_bin14
528 prop_binSR_electron2016M_bin2_ZZtoLep
529 prop_binCRTT_muon2017_bin4
530 prop_binCROS_muon2017_bin1_TTTo2L2Nu
531 prop_binCRTT_electron2016M_bin12
532 prop_binCRF_electron2017_bin2
533 prop_binCRTT_electron2017_bin9
534 prop_binCRF_electron2018_bin12_WrongSign
535 prop_binCROS_muon2017_bin1_VG
536 prop_binCRTT_electron2017_bin8
537 prop_binCRF_electron2016M_bin14
538 prop_binCRTT_muon2017_bin12
539 prop_binCRF_muon2018_bin7
540 prop_binCRTT_electron2018_bin18
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$

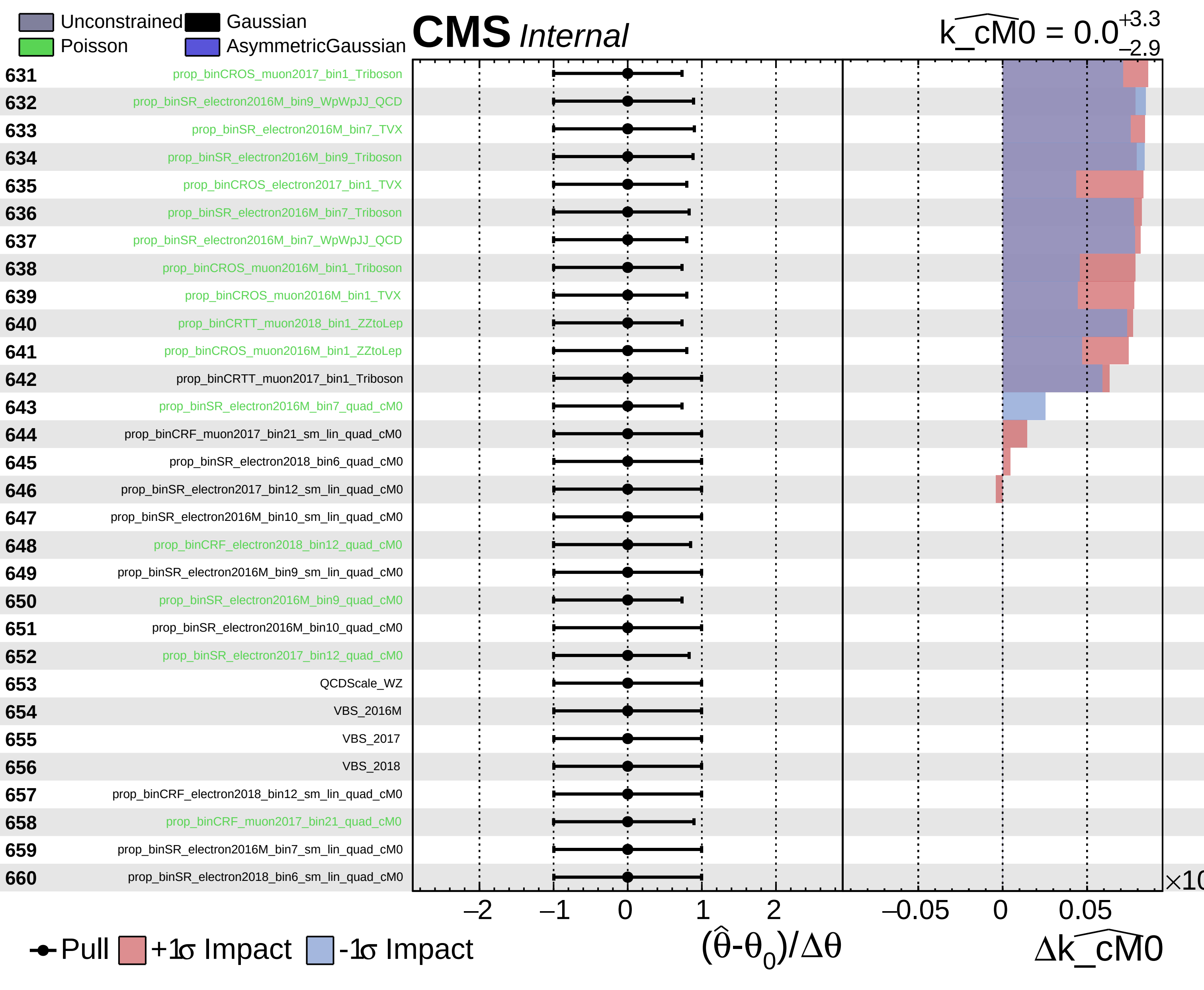


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
541 prop_binCRF_electron2017_bin7
542 prop_binCRF_electron2018_bin2
543 prop_binSR_muon2016M_bin2
544 prop_binCRTT_electron2016M_bin13
545 prop_binCRTT_electron2016M_bin19
546 prop_binSR_electron2016M_bin7_sm
547 prop_binCRTT_muon2017_bin11
548 prop_binCRTT_electron2017_bin13
549 prop_binCROS_electron2016M_bin13
550 prop_binCROS_muon2016M_bin17
551 prop_binCRTT_muon2016M_bin13
552 prop_binCROS_electron2016M_bin15
553 QCDScale_Triboson
554 prop_binCROS_muon2016M_bin1_TTTo2L2Nu
555 prop_binSR_electron2016M_bin4
556 prop_binCRF_electron2017_bin11
557 lep_2017
558 prop_binSR_electron2017_bin3
559 prop_binSR_electron2017_bin2
560 prop_binCRTT_muon2017_bin9
561 QCDScale_TVX
562 prop_binCRF_muon2017_bin21_WrongSign
563 prop_binCRTT_electron2018_bin16
564 prop_binSR_muon2018_bin3
565 prop_binCROS_electron2018_bin1_ZZtoLep
566 prop_binCRTT_muon2016M_bin8
567 prop_binCRTT_muon2018_bin21
568 prop_binCRTT_electron2016M_bin20
569 prop_binCROS_muon2016M_bin16
570 prop_binCROS_electron2017_bin1_TTTo2L2Nu
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM0

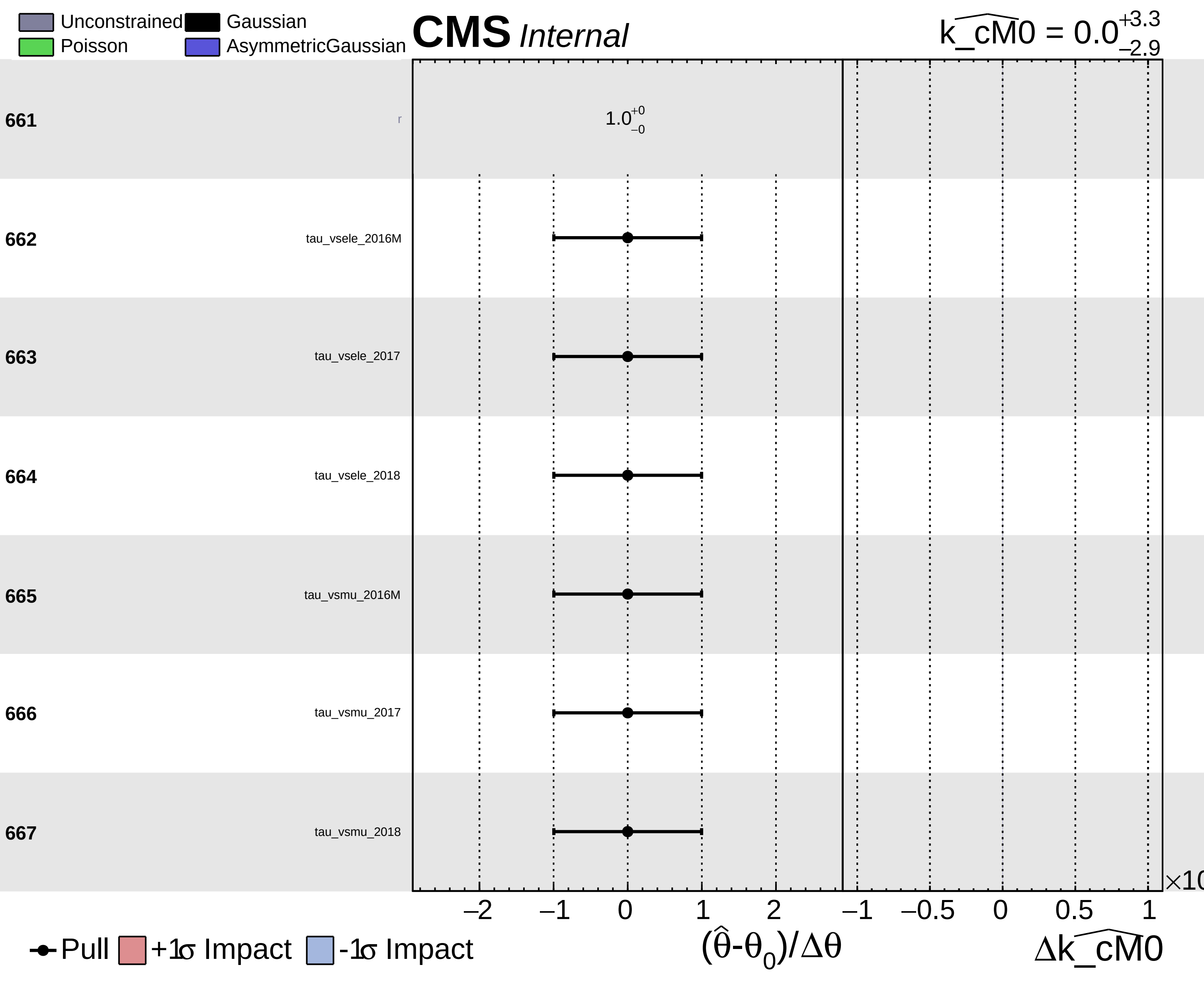

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
571 prop_binCROS_muon2016M_bin1_VG
572 prop_binCROS_electron2018_bin21
573 prop_binCRF_muon2017_bin21_WpWpJJ_QCD
574 prop_binCRF_electron2017_bin9
575 prop_binCRTT_muon2017_bin10
576 prop_binCROS_muon2017_bin1_TVX
577 prop_binCRTT_muon2016M_bin12
578 prop_binCRTT_muon2017_bin14
579 prop_binCROS_electron2016M_bin14_WZ
580 prop_binCROS_electron2016M_bin14_VG
581 prop_binCRTT_electron2017_bin15
582 prop_binCROS_electron2018_bin1_Triboson
583 prop_binSR_muon2016M_bin6
584 prop_binCROS_electron2016M_bin14_TVX
585 prop_binCRTT_electron2016M_bin8
586 prop_binCRTT_muon2016M_bin7
587 prop_binCRF_muon2017_bin21_ZZtoLep
588 prop_binCRF_muon2017_bin21_Triboson
589 prop_binCROS_electron2017_bin1_ZZtoLep
590 QCDScale_ZZtoLep
591 prop_binCROS_electron2016M_bin14_ZZtoLep
592 prop_binSR_electron2017_bin12_TTTo2L2Nu
593 prop_binSR_electron2016M_bin10_WZ
594 prop_binSR_electron2016M_bin7_WZ
595 prop_binCRF_electron2018_bin12_TTTo2L2Nu
596 prop_binCROS_electron2018_bin1_TVX
597 prop_binCROS_muon2016M_bin21
598 prop_binCRTT_muon2018_bin1_TVX
599 prop_binCROS_electron2016M_bin14_Triboson
600 prop_binSR_electron2016M_bin7_TTTo2L2Nu
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM0 = 0.0 +3.3 −2.9
601 prop_binCRTT_electron2018_bin1_TVX
602 prop_binCRTT_muon2017_bin1_TVX
603 prop_binCROS_muon2017_bin1_ZZtoLep
604 prop_binCRF_electron2018_bin12_ZZtoLep
605 prop_binSR_electron2017_bin12_ZZtoLep
606 prop_binSR_electron2016M_bin7_VG
607 prop_binCROS_electron2016M_bin1_Triboson
608 prop_binSR_electron2016M_bin10_TTTo2L2Nu
609 prop_binSR_electron2016M_bin9_VG
610 prop_binSR_electron2018_bin6_TTTo2L2Nu
611 prop_binSR_electron2016M_bin9_ZZtoLep
612 prop_binCRTT_muon2018_bin1_Triboson
613 prop_binSR_electron2016M_bin7_ZZtoLep
614 prop_binCROS_electron2016M_bin1_ZZtoLep
615 prop_binSR_electron2017_bin12_TVX
616 prop_binCRF_electron2018_bin12_TVX
617 prop_binSR_electron2018_bin6_ZZtoLep
618 prop_binSR_electron2016M_bin9_TVX
619 prop_binSR_electron2016M_bin10_Triboson
620 prop_binSR_electron2018_bin6_TVX
621 prop_binSR_electron2016M_bin10_TVX
622 FES
623 prop_binSR_electron2016M_bin10_ZZtoLep
624 prop_binSR_electron2016M_bin10_WpWpJJ_QCD
625 prop_binCRF_electron2018_bin12_Triboson
626 prop_binCRF_electron2018_bin12_WpWpJJ_QCD
627 prop_binSR_electron2017_bin12_WpWpJJ_QCD
628 prop_binSR_electron2017_bin12_Triboson
629 prop_binSR_electron2018_bin6_Triboson
630 prop_binSR_electron2018_bin6_WpWpJJ_QCD
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
631 prop_binCROS_muon2017_bin1_Triboson
632 prop_binSR_electron2016M_bin9_WpWpJJ_QCD
633 prop_binSR_electron2016M_bin7_TVX
634 prop_binSR_electron2016M_bin9_Triboson
635 prop_binCROS_electron2017_bin1_TVX
636 prop_binSR_electron2016M_bin7_Triboson
637 prop_binSR_electron2016M_bin7_WpWpJJ_QCD
638 prop_binCROS_muon2016M_bin1_Triboson
639 prop_binCROS_muon2016M_bin1_TVX
640 prop_binCRTT_muon2018_bin1_ZZtoLep
641 prop_binCROS_muon2016M_bin1_ZZtoLep
642 prop_binCRTT_muon2017_bin1_Triboson
643 prop_binSR_electron2016M_bin7_quad_cM0
644 prop_binCRF_muon2017_bin21_sm_lin_quad_cM0
645 prop_binSR_electron2018_bin6_quad_cM0
646 prop_binSR_electron2017_bin12_sm_lin_quad_cM0
647 prop_binSR_electron2016M_bin10_sm_lin_quad_cM0
648 prop_binCRF_electron2018_bin12_quad_cM0
649 prop_binSR_electron2016M_bin9_sm_lin_quad_cM0
650 prop_binSR_electron2016M_bin9_quad_cM0
651 prop_binSR_electron2016M_bin10_quad_cM0
652 prop_binSR_electron2017_bin12_quad_cM0
653 QCDScale_WZ
654 VBS_2016M
655 VBS_2017
656 VBS_2018
657 prop_binCRF_electron2018_bin12_sm_lin_quad_cM0
658 prop_binCRF_muon2017_bin21_quad_cM0
659 prop_binSR_electron2016M_bin7_sm_lin_quad_cM0
660 prop_binSR_electron2018_bin6_sm_lin_quad_cM0
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM0} = 0.0^{+3.3}_{-2.9}$
661
r
1.0^{+0}_{-0}
662
tau_vsele_2016M
663
tau_vsele_2017
664
tau_vsele_2018
665
tau_vsmu_2016M
666
tau_vsmu_2017
667
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM0}$